THOMAS MORE | UNIVERSITY OF APPLIED SCIENCES

Data vault

Agenda

- Introduction: General Context
- Why Data Vault (versus 3NF and Dimensional Model)
- What is a Data Vault
  - Official Definition (Dan Linstedt)
  - Difference with other data modelling techniques
  - A little bit of History
  - Evolution towards DV
- Data Vault Basics
  - Main Guidelines
  - Hub, Link and Satellites
- Business Vault
  - Purpose
  - Point-In-Time (PIT)
  - Bridge Tables
  - Predefined Calculations
- Conclusions
- Assignment: Thomas More Airport Survey

Intro

Modern Times of Big Data trends (AI, ML, Data Engineers, Agile …)
vs.
Reporting and data analysis needs (traditional Business Intelligence)

Traditional "analysis / development" track (month – years)
vs.
Changing source systems (rework)

Data Vault:
Methodology to model and develop EDW in … way:
-Agile
-Flexible
-Scalable

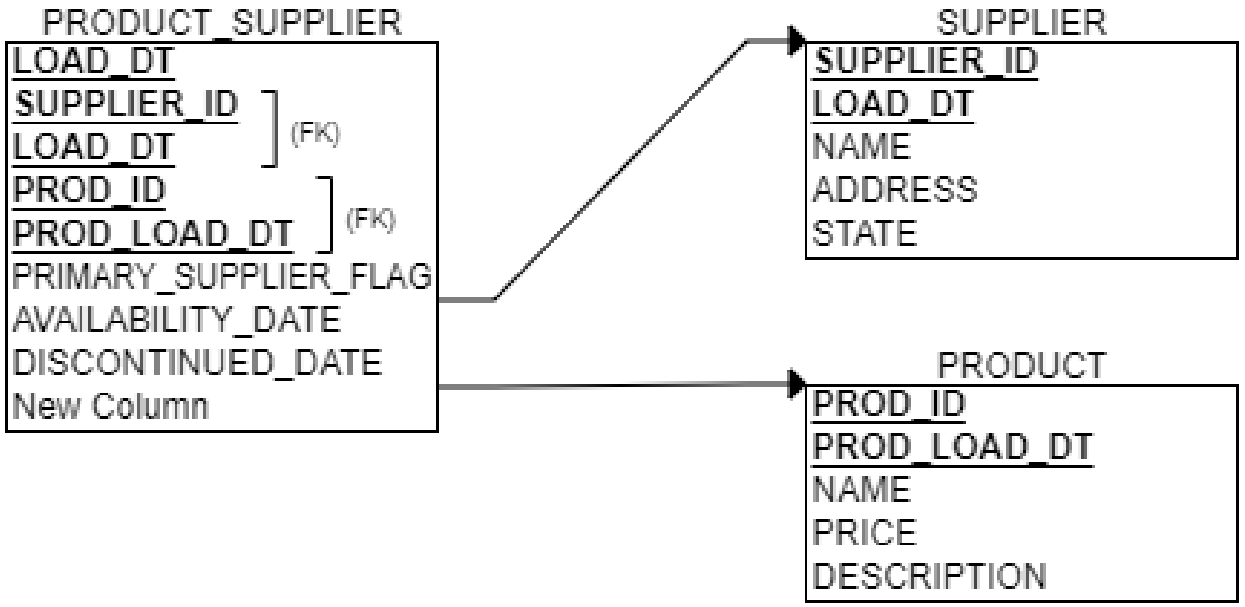
Agenda

- Why Data Vault (versus 3NF and Dimensional Model)

Why Data vault

(3NF)

- Where do we come from and what are the issues
  - 3NF
    - History Collection: need for a TIMESTAMP on every table
    - PK: Functional PK + Timestamp ← avoid duplicates
    - <u>Issues:</u>
      - CDC for loading
      - Complicated SQL queries for reading
      - Change of Initial Model
        - Adding new FK → reload table
        - Potential loss of History



Why Data vault

(3NF example)

Check the PRODUCT_SUPPLIER primary key

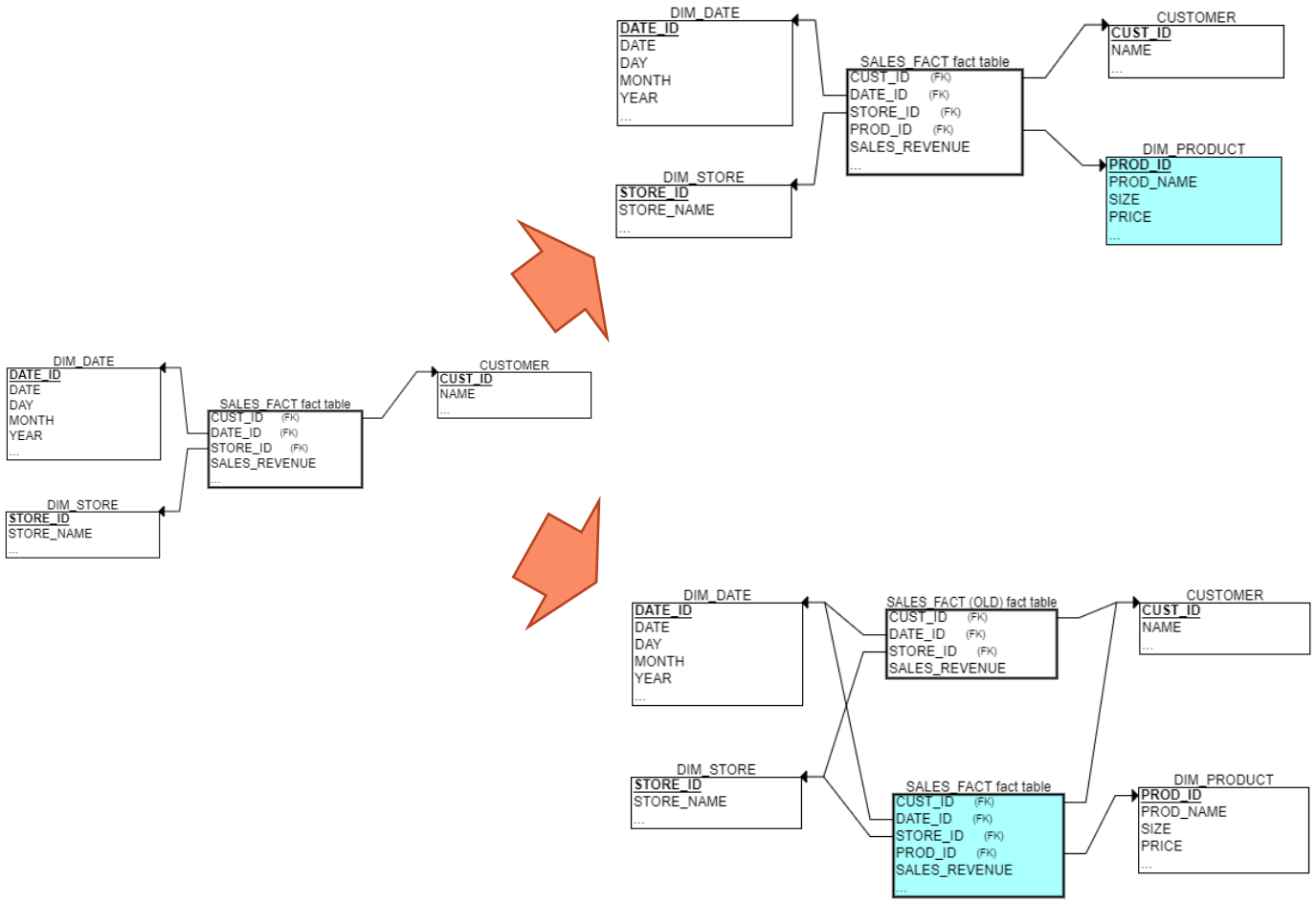
Imagine a MASTER-DETAIL-DETAIL

Why Data vault

(Dimensional Model)

Where do we come from and what are the issues

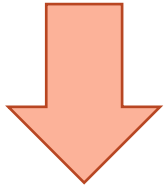
- Dimensional Model
  - New Requirements
  - Change in Level of Granularity of FACT
  - Issues:
    - Rebuilding Aggregated Fact
      - Reload all the data from the source
      - Possibly loss of history
    - Build new Aggregated Fact
      - Redundant facts
      - Possibly missing history in the source



Why Data vault (Dimensional Model)

- FACT aggregation level changed
- Old History isn't valid anymore
- Possible loss of history if not available in the source systems

Imagine a DWH with 100+ tables and 1M+ rows
Refactoring and reloading will cost a LOT of time and effort



DATA VAULT modeling method gives us an
AGILE DATA-ENIGEERING APPROACH to avoid those issue

WHAT IS A DATA VAULT

- Official Definition (Dan Linstedt)
- Difference with other data modelling techniques
- A little bit of History
- Evolution towards DV

Official Definition from Dan Linstedt

The Data Vault is a **detail oriented**, **historical tracking** and **uniquely linked** set of **normalized tables** that support **one or more functional areas** of business.

It is a **hybrid approach** encompassing the best of breed between **3rd normal form** (3NF) and **star schema**. The design is **flexible**, **scalable**, **consistent**, and **adaptable** to the needs of the enterprise. It is a data model that is architected specifically to meet the needs of today's enterprise data warehouses.

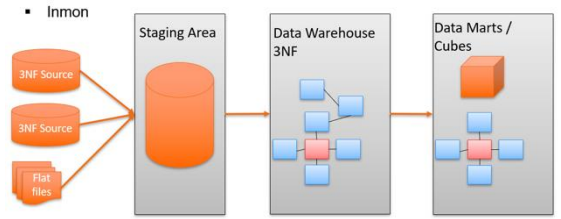
The main point here is that Data Vault (DV) was developed specifically to address the issues in the data warehousing space. It was built to be a **non-volatile**, **auditable**, **historical** repository of **enterprise data**.
It does not replace star schemas needed at the reporting layer

Three KEY FEATURES:

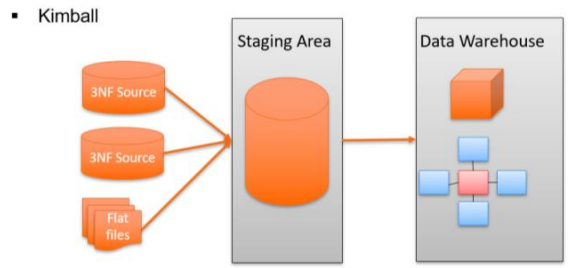
- **Flexibility**: rapid change on massive scale without impact
- **Productivity**: low complexity with high value output at rapid pace
- **Scalability**: no foreseeable barrier to increased size and scope

WHERE does it fit in the DWH architecture

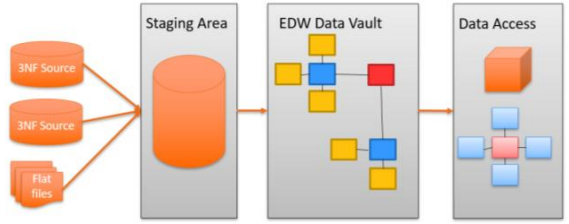
Compared to:
INMON: the Data Vault fits between the STAGE layer and DATAMART layer



KIMBALL: the Data Vault becomes the PERSISTENT STAGING AREA that feeds the DIMENSIONAL DATA WAREHOUSE

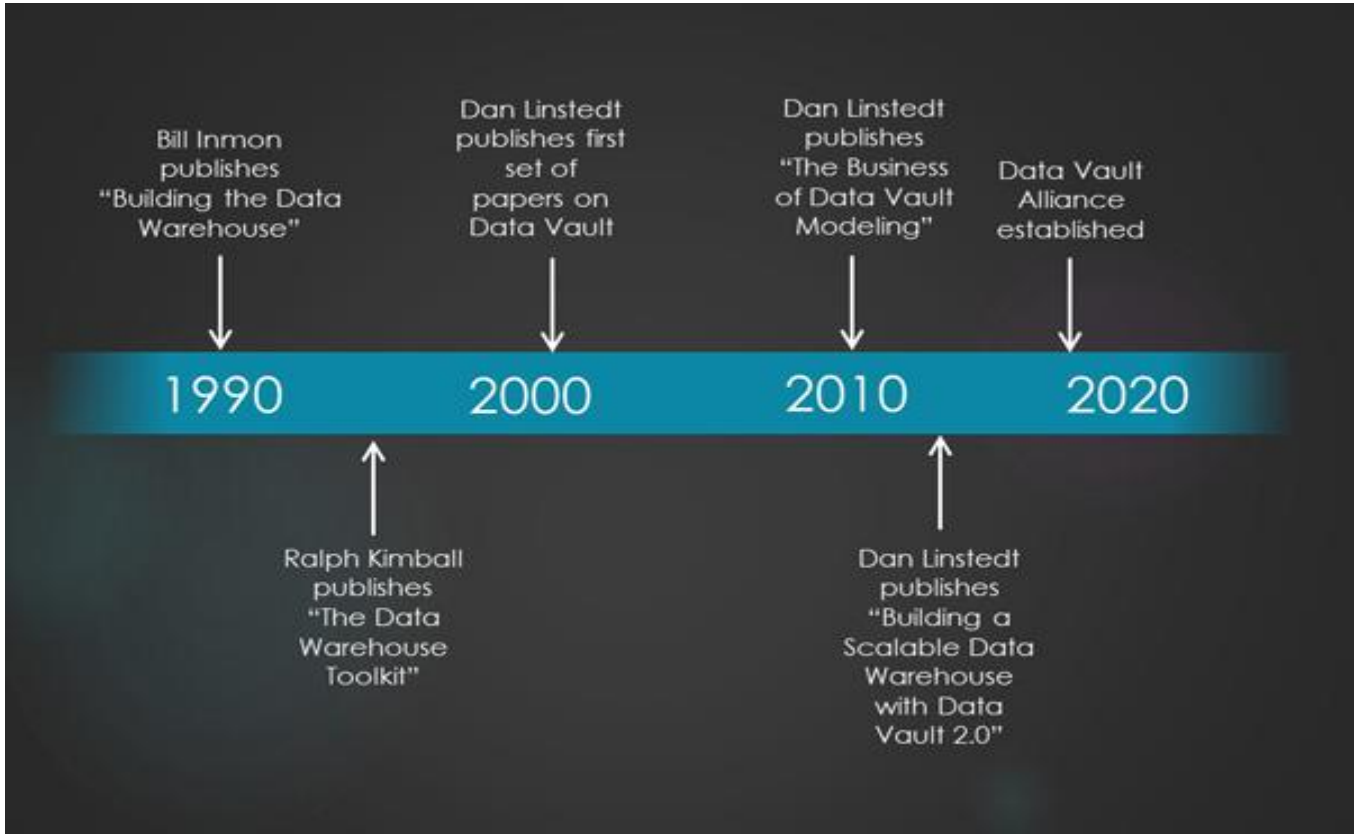


DATA VAULT: the concept is not new, but the modeling technique is

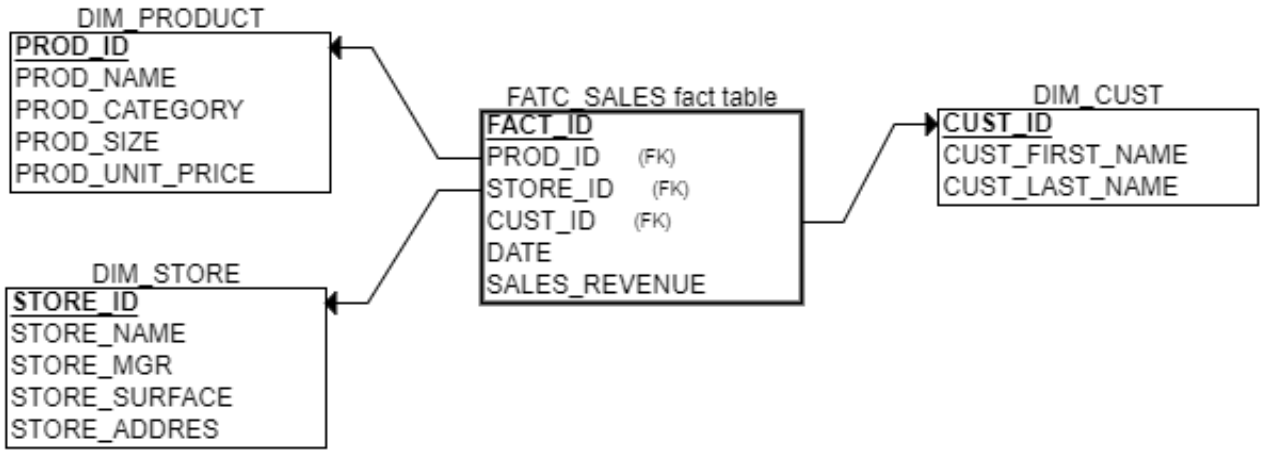


A little bit of history

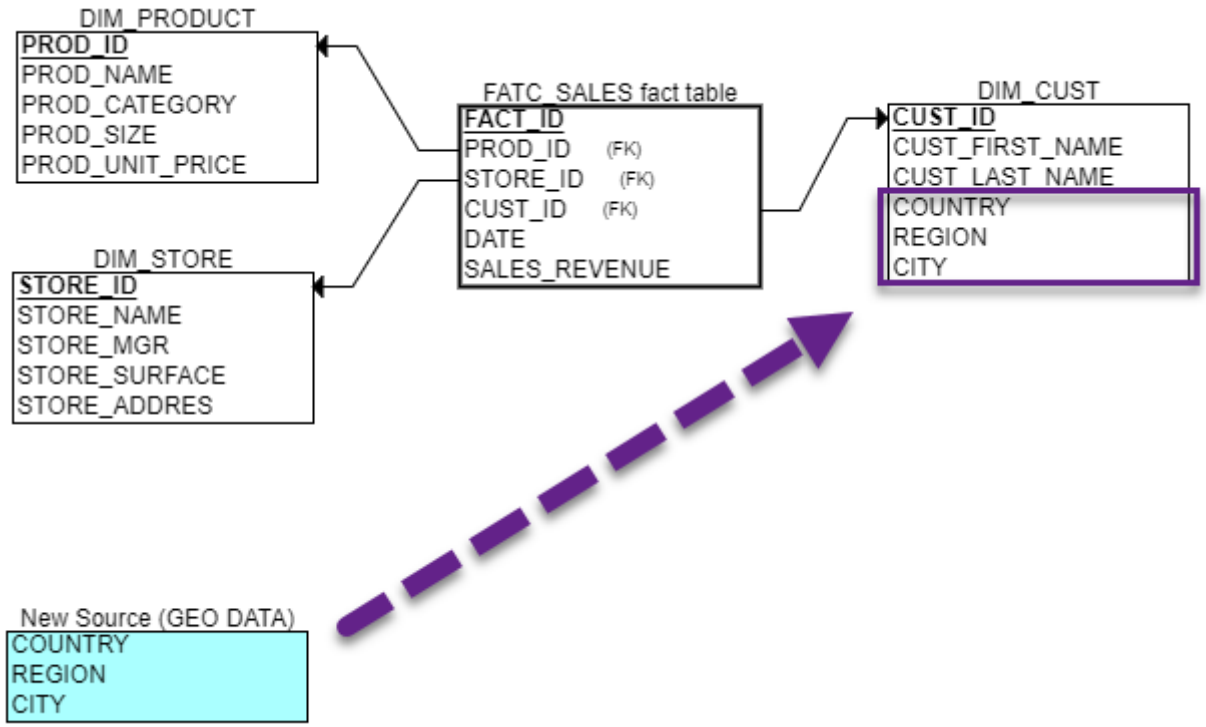
- Origins of DV go back to 1990's, developed by Dan Linstedt
- Refined in 2010, he published his first book
- In 2008 Bill Inmon is quoted as supporting Data Vault as the method for his new DW 2.0 approach.



EVOLUTION FROM DIMENSIONAL MODELING TOWARDS DATA VAULT

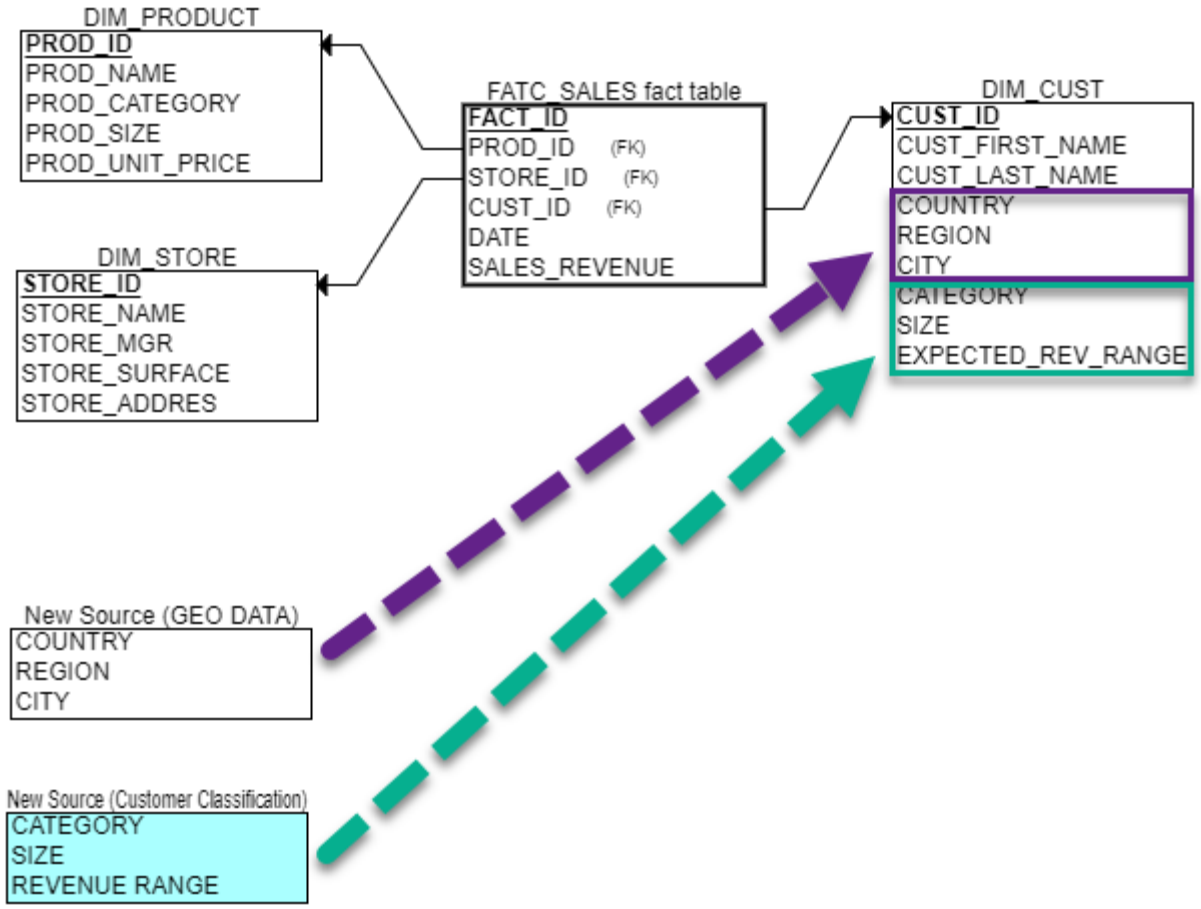


WE HAVE A STAR SCHEMA, BASED ON EXISTING REQUIREMENTS



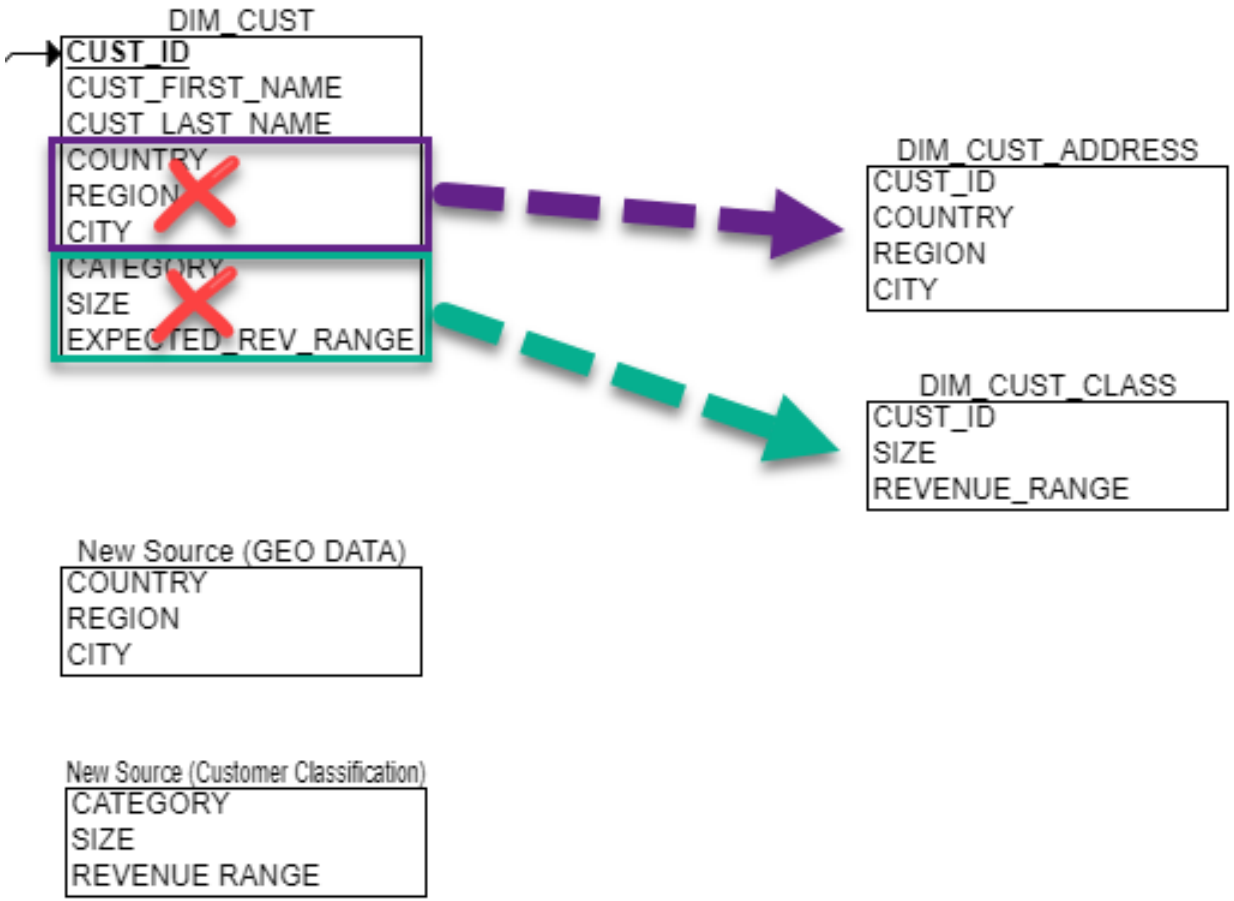
Evolution from Dimensional Modeling towards DATA vault

BUT … THE SOURCE SYSTEM RECEIVED A MAKE-OVER, AND GEO DATA HAS BEEN ADDED
AND DIM_CUST MUST BE REWORKED



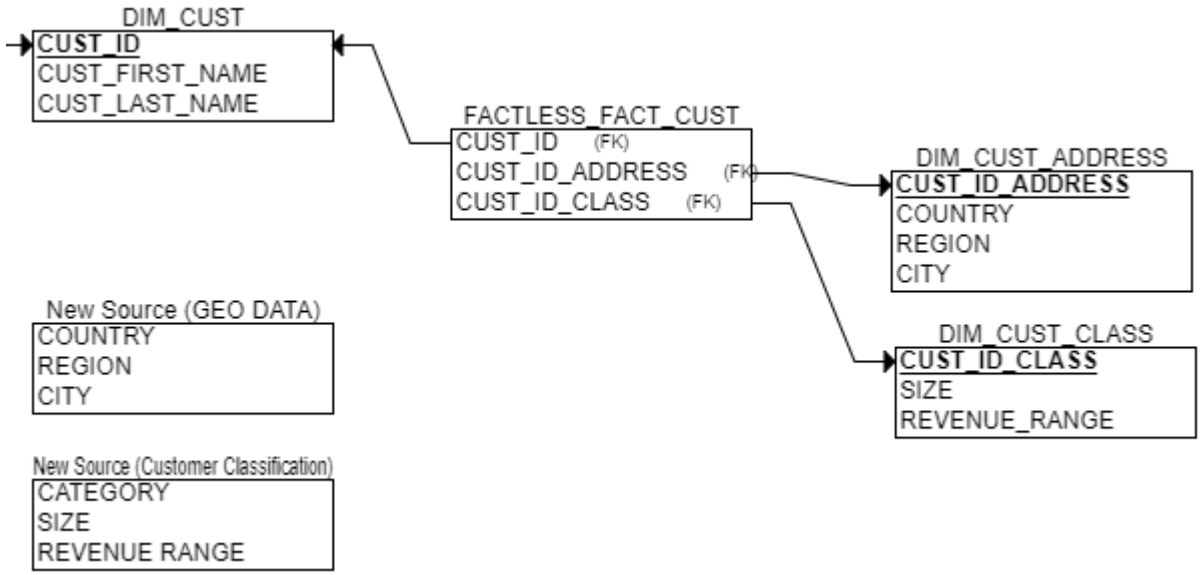
Evolution from Dimensional Modeling towards DATA vault

A NEW SOURCE SYSTEM WAS PURCHASED TO ADD A CUSTOMER CLASSIFICATION FOR THE MARKETING TEAM AGAIN, REWORK



Evolution from Dimensional Modeling towards DATA vault

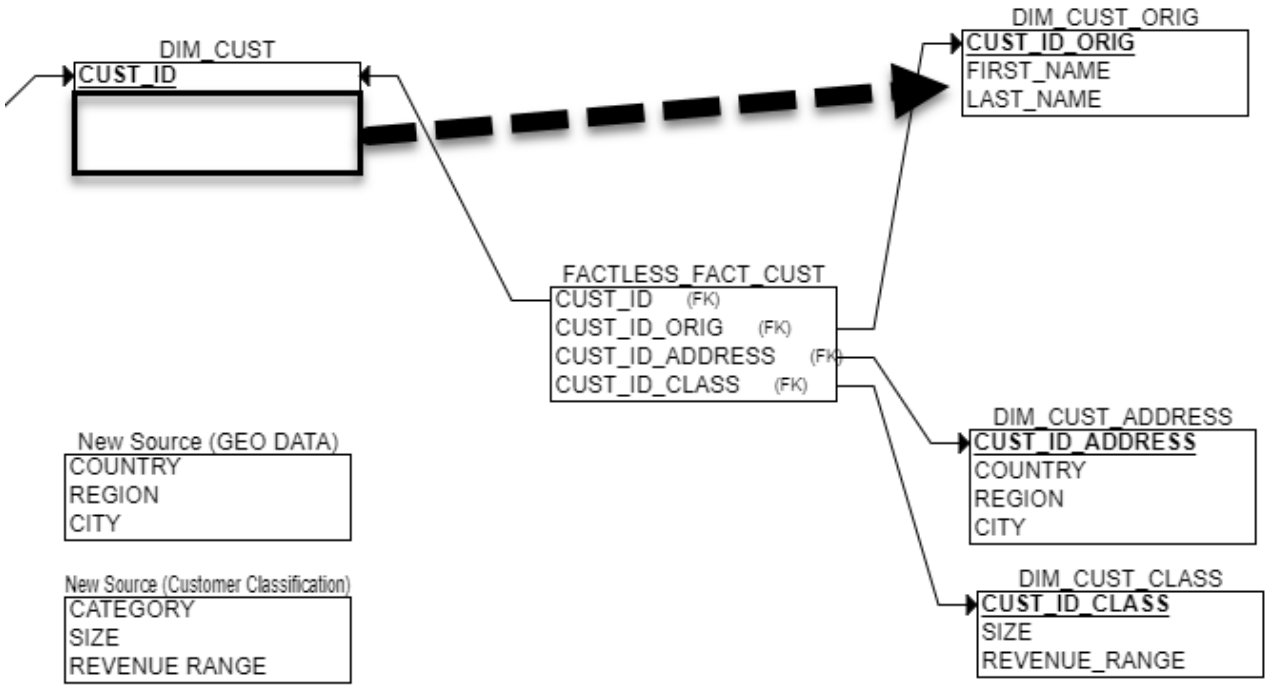
IN DIMENSIONAL MODELING, YOU WOULD WANT TO SPLIT THESE OUT



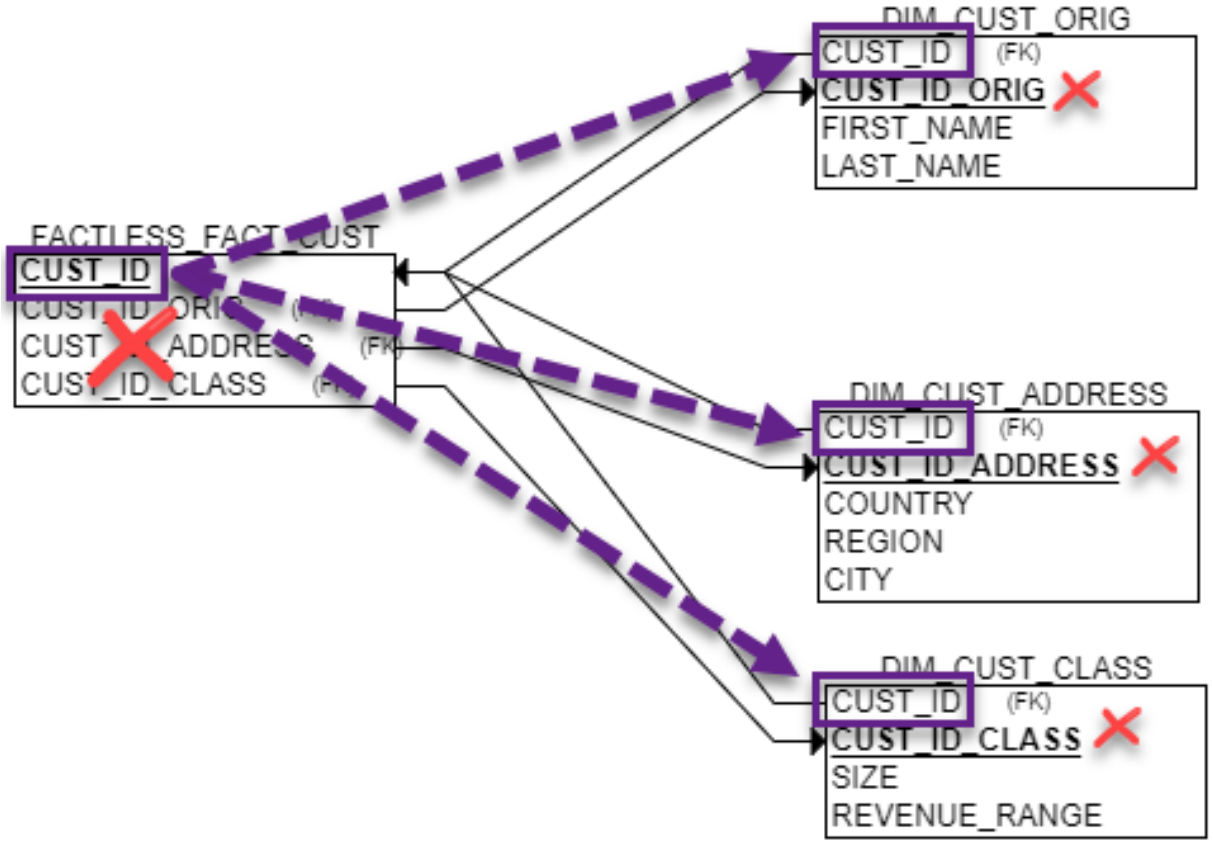
Evolution from Dimensional Modeling towards DATA vault

ALL THESE MINI-DIMENSIONS, WITH EACH HAVING ITS OWN CUSTOMER CHARACTERISTICS, ARE THEN LINK TOGETHER VIA A FACT-LESS FACT, TO GUARANTEE CONFORMITY

Evolution from Dimensional Modeling towards DATA vault



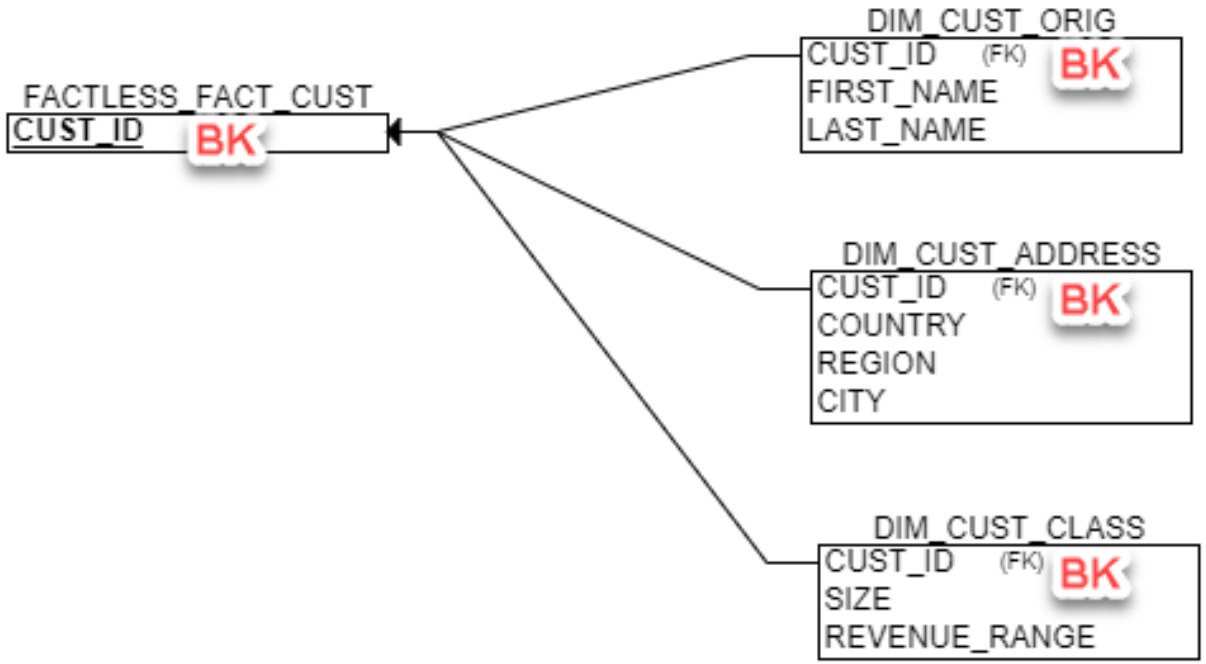
NEXT STEP IN THE EVOLUTION IS TO PULL OUT THE ORIGINAL CUSTOMER CHARACTERISTICS INTO ANOTHER MINI-DIMENSION, KEEPING ONLY THE ORIGINAL CUSTOMER BUSINESS KEY



Evolution from Dimensional Modeling towards DATA vault

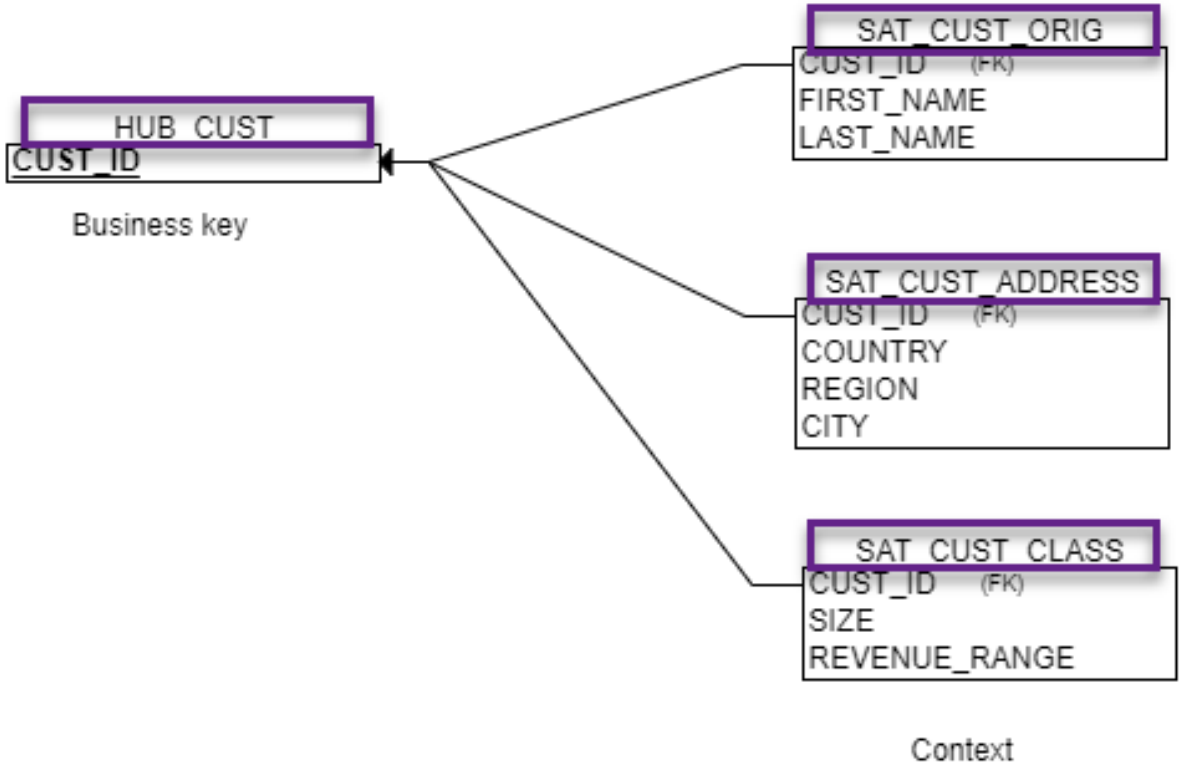
REVERSE ALL RELATIONSHIPS, AND MIGRATE THE BUSINESS KEY OUT INSTEAD OF IN, AND REMOVE ALL THE OTHER KEYS

Evolution from Dimensional Modeling towards DATA vault



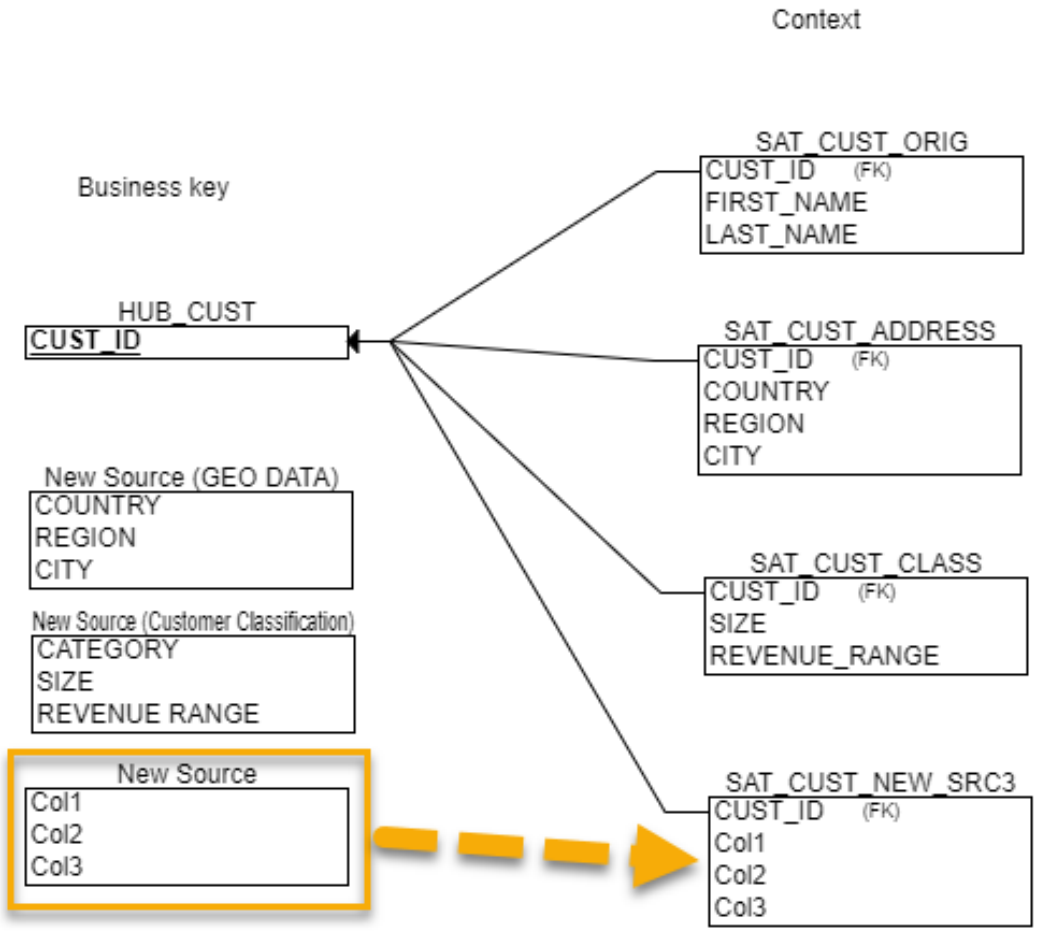
SO WE END UP WITH JUST THE BUSINESS KEY, AND MULTIPLE CONTEXTS FOR THAT BUSINESS KEY

Evolution from Dimensional Modeling towards DATA vault



THIS EVOLUTION ENDED UP IN A HUB CUSTOMER (DEFINED BY BK) AND VARIOUS SATELLITES FOR THE CONTEXTS REFERRING TO THE SAME BK

Evolution from Dimensional Modeling towards DATA vault



NEW SOURCE WITH NEW CUSTOMER CONTEXT DATA, RESULTING IN A NEW SAT.

NO REWORK NEEDED OF THE EXISTING STRUCTURE.

Agenda

Data Vault Basics

- Main Guidelines
- Hub
- Link
- Satellites

Main GUIDELINES:

- Load data as it exists in the source: no edits, no changes, no data cleansing, no application of soft business rules
- Reason:
  - Auditability: trace data to the source
  - History: once the source has deleted historical data, the EDW remains the Source Of Record.
- DV is a source of FACTS, not a source of TRUTH

Hard vs SOFT Business rules:

Hard Rules

These should be applied before data is stored in the Data Vault. Any rules applied here do **not alter** the contents or the granularity of the data, and **maintains auditability**.

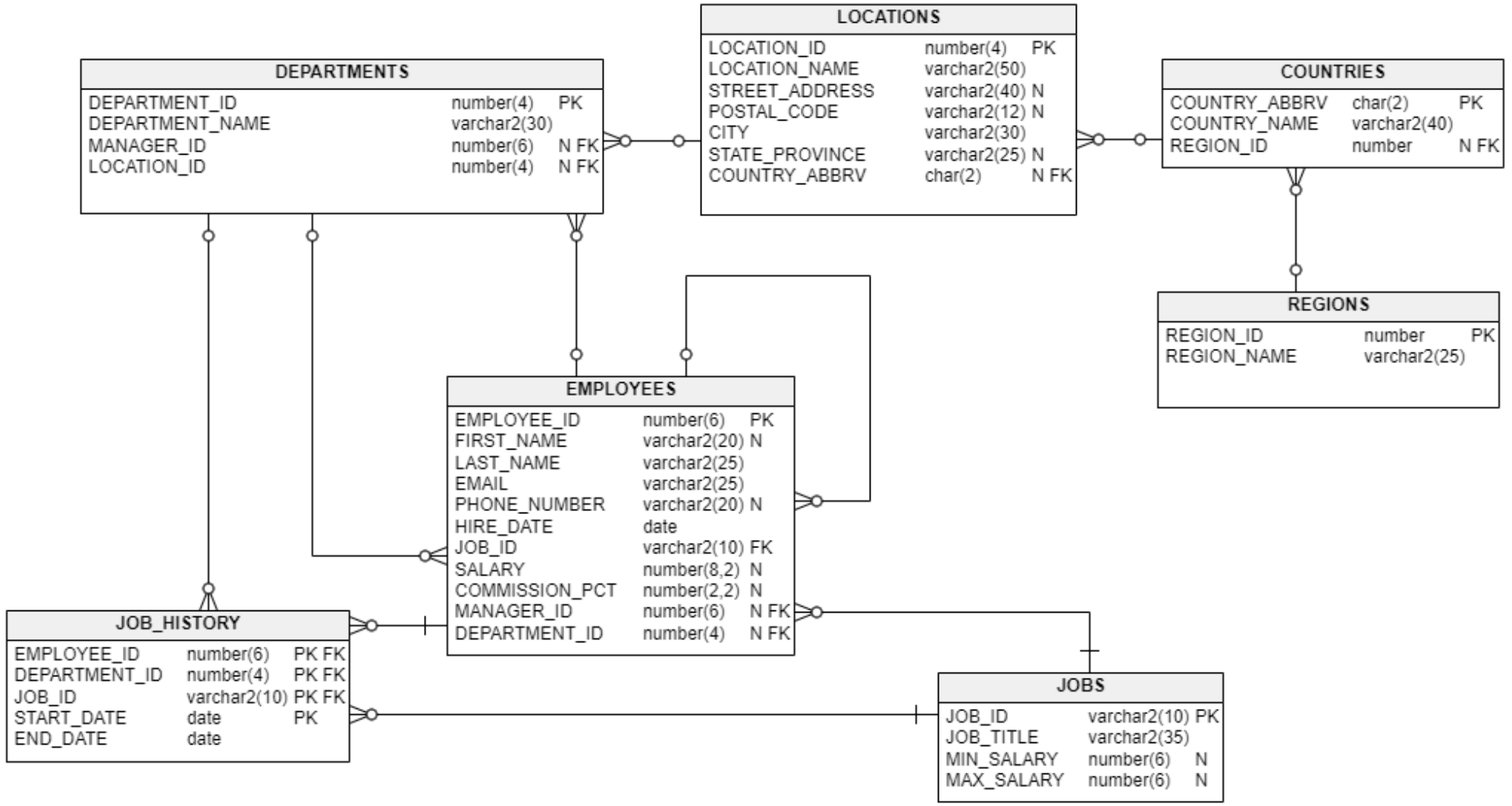
- Data typing
- Normalization / Denormalization
- Adding system fields (tags)
- De-duplication
- Splitting by record structure
- Trimming spaces from character strings

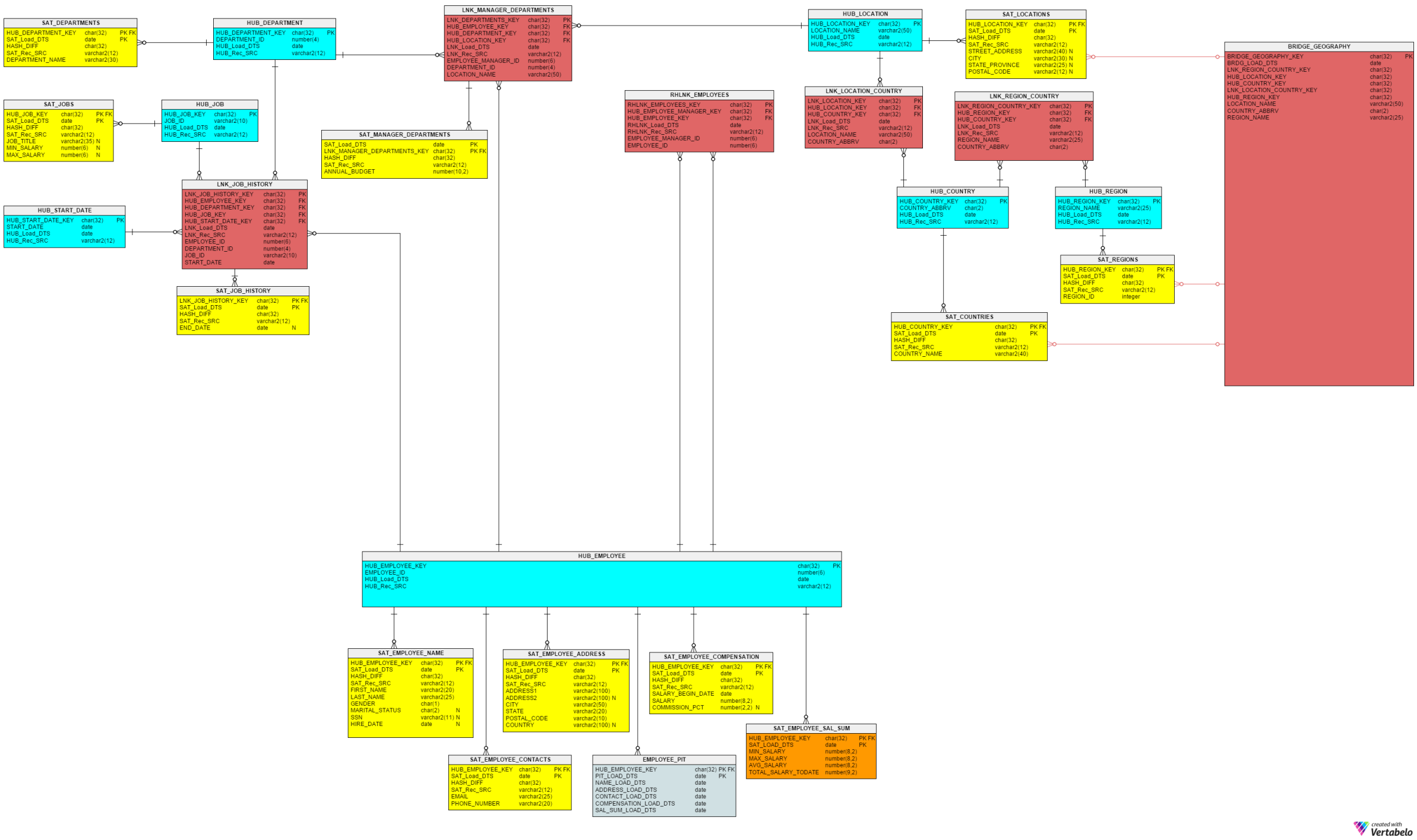
Soft Rules

Rules that change, or interpret the data, for example adds business logic. This changes the granularity of the data.

- Concatenating name fields
- Standardizing addresses
- Calculating monthly sales
- Coalescing / decoding
- Consolidation

USED SOURCE SYSTEM

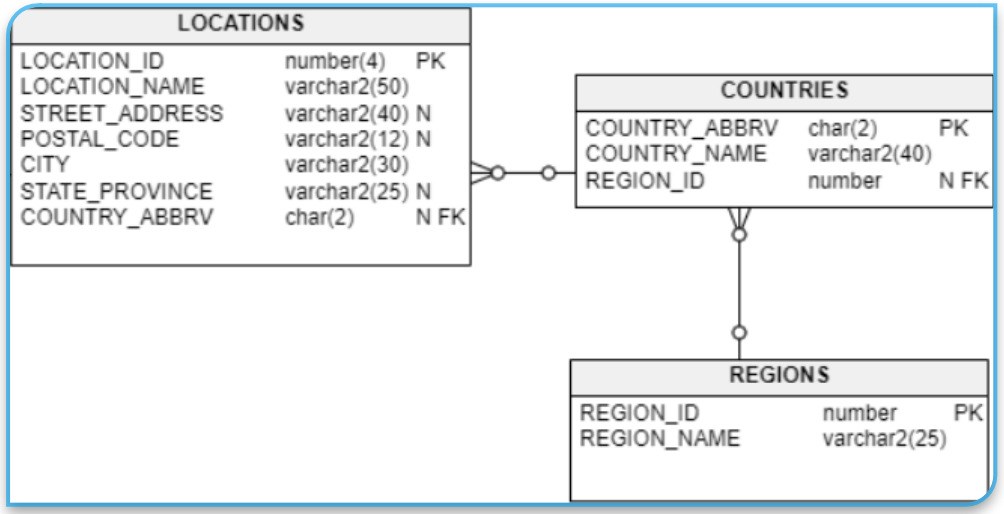



SAT_DEPARTMENTS
HUB_DEPARTMENT_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
DEPARTMENT_NAME varchar2(30)
HUB_DEPARTMENT
HUB_DEPARTMENT_KEY char(32) PK
DEPARTMENT_ID number(4)
HUB_Load_DTS date
HUB_Rec_SRC varchar2(12)
LNK_MANAGER_DEPARTMENTS
LNK_DEPARTMENTS_KEY char(32) PK
HUB_EMPLOYEE_KEY char(32) FK
HUB_DEPARTMENT_KEY char(32) FK
HUB_LOCATION_KEY char(32) FK
LNK_Load_DTS date
LNK_Rec_SRC varchar2(12)
EMPLOYEE_MANAGER_ID number(6)
DEPARTMENT_ID number(4)
LOCATION_NAME varchar2(50)
HUB_LOCATION
HUB_LOCATION_KEY char(32) PK
LOCATION_NAME varchar2(50)
HUB_Load_DTS date
HUB_Rec_SRC varchar2(12)
SAT_LOCATIONS
HUB_LOCATION_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
STREET_ADDRESS varchar2(40) N
CITY varchar2(30) N
STATE_PROVINCE varchar2(25) N
POSTAL_CODE varchar2(12) N
BRIDGE_GEOGRAPHY
BRIDGE_GEOGRAPHY_KEY char(32) PK
BRDG_LOAD_DTS date
LNK_REGION_COUNTRY_KEY char(32)
HUB_LOCATION_KEY char(32)
HUB_COUNTRY_KEY char(32)
LNK_LOCATION_COUNTRY_KEY char(32)
HUB_REGION_KEY char(32)
LOCATION_NAME varchar2(50)
COUNTRY_ABBRV char(2)
REGION_NAME varchar2(25)
SAT_JOBS
HUB_JOB_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
JOB_TITLE varchar2(35) N
MIN_SALARY number(6) N
MAX_SALARY number(6) N
HUB_JOB
HUB_JOB_KEY char(32) PK
JOB_ID varchar2(10)
HUB_Load_DTS date
HUB_Rec_SRC varchar2(12)
SAT_MANAGER_DEPARTMENTS
SAT_Load_DTS date PK
LNK_MANAGER_DEPARTMENTS_KEY char(32) PK FK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
ANNUAL_BUDGET number(10,2)
RHLNK_EMPLOYEES
RHLNK_EMPLOYEES_KEY char(32) PK
HUB_EMPLOYEE_MANAGER_KEY char(32) FK
HUB_EMPLOYEE_KEY char(32) FK
RHLNK_Load_DTS date
RHLNK_Rec_SRC varchar2(12)
EMPLOYEE_MANAGER_ID number(6)
EMPLOYEE_ID number(6)
LNK_LOCATION_COUNTRY
LNK_LOCATION_KEY char(32) PK
HUB_LOCATION_KEY char(32) FK
HUB_COUNTRY_KEY char(32) FK
LNK_Load_DTS date
LNK_Rec_SRC varchar2(12)
LOCATION_NAME varchar2(50)
COUNTRY_ABBRV char(2)
LNK_REGION_COUNTRY
LNK_REGION_COUNTRY_KEY char(32) PK
HUB_REGION_KEY char(32) FK
HUB_COUNTRY_KEY char(32) FK
LNK_Load_DTS date
LNK_Rec_SRC varchar2(12)
REGION_NAME varchar2(25)
COUNTRY_ABBRV char(2)
LNK_JOB_HISTORY
LNK_JOB_HISTORY_KEY char(32) PK
HUB_EMPLOYEE_KEY char(32) FK
HUB_DEPARTMENT_KEY char(32) FK
HUB_JOB_KEY char(32) FK
HUB_START_DATE_KEY char(32) FK
LNK_Load_DTS date
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EMPLOYEE_ID number(6)
DEPARTMENT_ID number(4)
JOB_ID varchar2(10)
START_DATE date
HUB_START_DATE
HUB_START_DATE_KEY char(32) PK
START_DATE date
HUB_Load_DTS date
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HUB_COUNTRY
HUB_COUNTRY_KEY char(32) PK
COUNTRY_ABBRV char(2)
HUB_Load_DTS date
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HUB_REGION
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SAT_Rec_SRC varchar2(12)
REGION_ID integer
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LNK_JOB_HISTORY_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
END_DATE date N
SAT_COUNTRIES
HUB_COUNTRY_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
COUNTRY_NAME varchar2(40)
HUB_EMPLOYEE
HUB_EMPLOYEE_KEY char(32) PK
EMPLOYEE_ID number(6)
HUB_Load_DTS date
HUB_Rec_SRC varchar2(12)
SAT_EMPLOYEE_NAME
HUB_EMPLOYEE_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
FIRST_NAME varchar2(20)
LAST_NAME varchar2(25)
GENDER char(1)
MARITAL_STATUS char(2) N
SSN varchar2(11) N
HIRE_DATE date N
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HUB_EMPLOYEE_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
ADDRESS1 varchar2(100)
ADDRESS2 varchar2(100) N
CITY varchar2(50)
STATE varchar2(20)
POSTAL_CODE varchar2(10)
COUNTRY varchar2(100) N
SAT_EMPLOYEE_COMPENSATION
HUB_EMPLOYEE_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
SALARY_BEGIN_DATE date
SALARY number(8,2)
COMMISSION_PCT number(2,2) N
SAT_EMPLOYEE_SAL_SUM
HUB_EMPLOYEE_KEY char(32) PK FK
SAT_LOAD_DTS date PK
MIN_SALARY number(8,2)
MAX_SALARY number(8,2)
AVG_SALARY number(8,2)
TOTAL_SALARY_TODATE number(9,2)
SAT_EMPLOYEE_CONTACTS
HUB_EMPLOYEE_KEY char(32) PK FK
SAT_Load_DTS date PK
HASH_DIFF char(32)
SAT_Rec_SRC varchar2(12)
EMAIL varchar2(25)
PHONE_NUMBER varchar2(20)
EMPLOYEE_PIT
HUB_EMPLOYEE_KEY char(32) PK FK
PIT_LOAD_DTS date PK
NAME_LOAD_DTS date
ADDRESS_LOAD_DTS date
CONTACT_LOAD_DTS date
COMPENSATION_LOAD_DTS date
SAL_SUM_LOAD_DTS date
created with Vertabelo

hubs

- It’s the first step, they are the core of the DV design
- Allow to integrate multiple source systems
- Are source system agnostic
- Are based on true business keys or meaningful natural keys. Identifiable business elements
  - Column(s) found in the systems that the business consistently uses to identify and locate the data
  - How:
    - alternate unique keys or indexes
    - data profiling and conversations with the business customers
- Don’t use source system surrogate keys

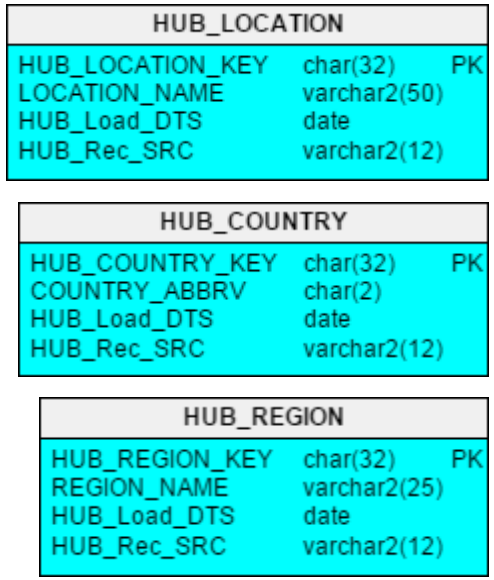
HUBS: find the correct business key



- PK: LOCATION_ID → Surrogate Key → NOK
  - Sequence (number): can be found anywhere else
  - No business meaning
- LOCATION_NAME → OK if:
  - Check with business
  - Check uniqueness in the table (profiling)
  - Look at indexed columns to help you
- COUNTRY_ABBREV and REGION_NAME will work for the other tables

Hubs

Definition



- List of unique business keys
- HUB Table Structure:
  - HUB PK: calculated field, HASH (SHA-265, MD5) of the Business Key Columns
    (No sequence or counter: issue with parallelism)
    ex: TRIM(UPPER(BK1))||'|'||TIM(UPPER(BK2))
    ex: SHA(STORE-BRUSSELS-WEST,265) :

    SHA256 Hash of your string:

    `240083C23394226B7D416CB8B50336F36E5F7C118FFE172BE4AFF2553D3AAC8A`
  - BUSINESS KEY COLUMN(s)
  - LOAD DATETIME: informs about the first load of a business key into the DWH. The row is never updated or dropped
  - RECORD SOURCE SYSTEM: informs about the first source system a business key came from
- HUBs are represented in the color BLUE

Hubs Example code

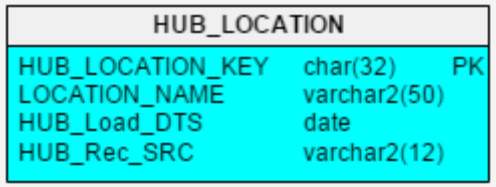
```
INSERT INTO VAULT.H_AIRPORT
(
        H_AIRPORT_SK
       ,H_AIRPORT_BK
       ,SOURCE_NAME
)
```

Where is the LOAD_DT ?

```
SELECT DISTINCT
       CAST(SHA2(COALESCE(ODS.FILE_ASQ_SURVEY.AIRPORT,'#MV'),256) AS VARCHAR(64))
                                                                    AS      H_AIRPORT_SK
       ,COALESCE(ODS.FILE_ASQ_SURVEY.AIRPORT,'#MV')                 AS      H_AIRPORT_BK
       ,'ASQ'                                                       AS      SOURCE_NAME
FROM ODS.FILE_ASQ_SURVEY
WHERE          1 = 1
       AND ERROR_IND = 0
       AND NOT EXISTS (
              SELECT 1 FROM VAULT.H_AIRPORT v
              WHERE v.H_AIRPORT_SK = CAST(SHA2(COALESCE(ODS.FILE_ASQ_SURVEY.AIRPORT,'#MV'),256) AS
VARCHAR(64))
       );
```

Hubs

what can you do with a single hub



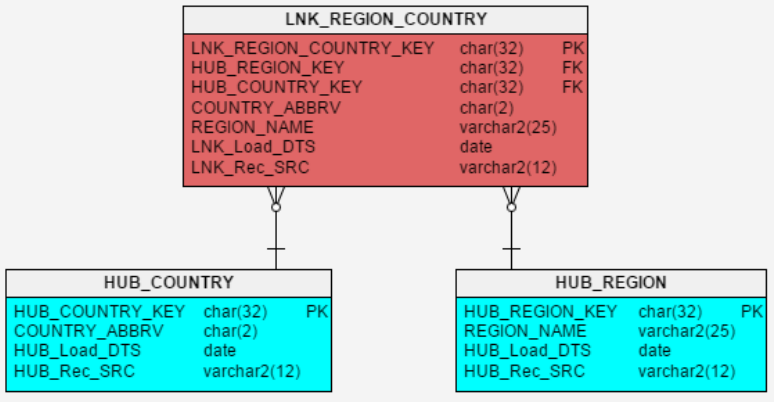
- Not much, since they only have the Business Key
- BUT , you can do Profiling and Basic Analysis on that BK :
  - How many Business Keys do we have?
  - How many source systems provide the data
  - Quality Checks:
    - Similar Business Keys from different sources

LINKS

- LINKs are the key to Flexibility and Scalability
  - Allow changes and additions to the model
  - Add new objects and relationships without having to change existing model and load scripts
- Represent Relationships and Events in the source
  - Relationship between hubs : COUNTRY – REGION
  - Events / transactions: SALES TRANSACTION involving CUSTOMER, PRODUCT, STORE and SALES DATE (revenue)
- As HUB's can have no FK's, the joins between HUB's are represented by LINK's
- The LINK captures the relationship of data elements at the lowest grain possible
- Union of Business Keys: it contains ALL the BK columns of the related HUB's (new since DV 2.0)

LINKs

Definition



- LINK Table Structure:
  - LINK PK: calculated field, HASH (SHA-265, MD5) of ALL the Business Key Columns of the contributing HUB's
    Ex: TRIM(UPPER(BK1))||'|'||TIM(UPPER(BK2))

    Ex: "BE-CENTER" :

    SHA256 Hash of your string:

    842C9AE428A9F5D0B856F7F4EB621ACE24F4EF1B8D6868F480FB115A0353104F
  - PK's from the contributing HUB's: used for lookup's
  - BUSINESS KEY COLUMN(s) from the HUB's (new since DV 2.0, for query performance)
  - LOAD DATETIME: informs about the first load of a combination of HUB business keys into the DWH. The row is never updated or dropped
  - RECORD SOURCE SYSTEM: informs about the first source system the link came from
- LINKs are represented in the color RED (or sometime GREEN)

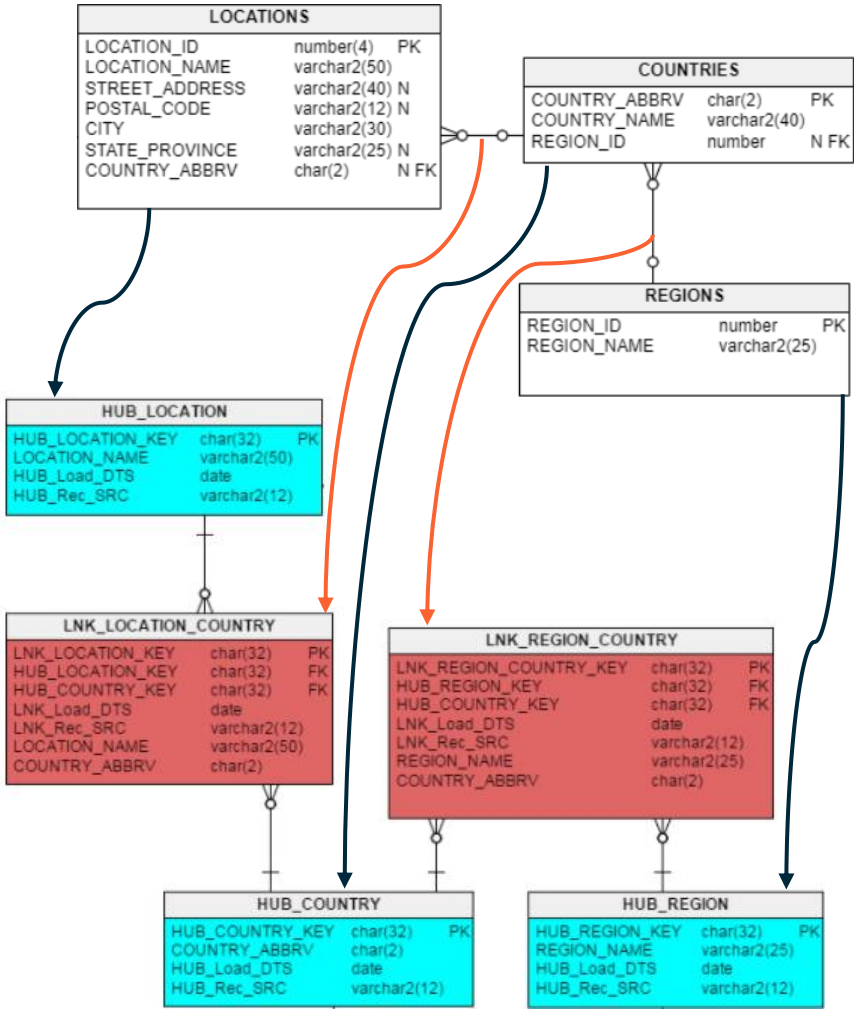
LINKs Example code

```
INSERT INTO VAULT.L_ASQ_QUESTION_ASQ_RESPONSIBLE
(
        L_ASQ_QUESTION_ASQ_RESPONSIBLE_SK
       ,H_ASQ_QUESTION_SK
       ,H_ASQ_RESPONSIBLE_SK
       ,SOURCE_NAME
)
```

Where is the LOAD_DT ?

```
SELECT DISTINCT
       CAST(SHA2(COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.DB_COLUMN_NAME,'#MV') || '|' ||
COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_NAME,'#MV') || '|' ||
COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_TYPE,'#MV'),256) AS VARCHAR(64))
                                                                        AS L_ASQ_QUESTION_ASQ_RESPONSIBLE_SK
       ,CAST(SHA2(COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.DB_COLUMN_NAME,'#MV'),256) AS VARCHAR(64))
                                                                        AS H_ASQ_QUESTION_SK
       ,CAST(SHA2(COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_NAME,'#MV') || '|' ||
COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_TYPE,'#MV'),256) AS VARCHAR(64))
                                                                        AS H_ASQ_RESPONSIBLE_SK
       ,'ASQ' AS SOURCE_NAME
FROM ODS.FILE_ASQ_QUESTION_RESPONSIBLE
```

```
WHERE 1 = 1
       AND NOT EXISTS (
                SELECT 1 FROM VAULT.L_ASQ_QUESTION_ASQ_RESPONSIBLE v
                WHERE v.L_ASQ_QUESTION_ASQ_RESPONSIBLE_SK =
CAST(SHA2(COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.DB_COLUMN_NAME,'#MV') || '|' ||
COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_NAME,'#MV') || '|' ||
COALESCE(ODS.FILE_ASQ_QUESTION_RESPONSIBLE.RESPONSIBLE_TYPE,'#MV'),256) AS VARCHAR(64))
       );
```



HUBs and LINKs

Source Entity used to define HUB
Join used to define LINK between HUBs

Remember: until here, we have only unique BK's and the relation between BK's

- No history
- No attributes

Satellites (SATs)

- SATs are where all the descriptive columns go, the non-key columns
- Here is where the CDC (Changed Data Capture) is done, and where the History is stored
  - The concept is very similar to a SCD2 (Slowly Changing Dimension Type 2)
- This is the only place in the DV that has a two-part PK:
  - The PK of the parent HUB or LINK
  - The LOAD_DT
- SAT's are Point-In-Time, they tell us what we did know when
- Never delete or update:
  - every time the DV is loaded, new records or changed records are inserted into the SATs with a timestamp

SAT Definition

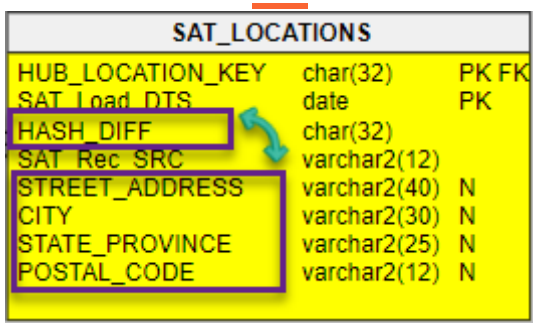
SAT_LOCATIONS		
HUB_LOCATION_KEY	char(32)	PK FK
SAT_Load_DTS	date	PK
HASH_DIFF	char(32)	
SAT_Rec_SRC	varchar2(12)	
STREET_ADDRESS	varchar2(40)	N
CITY	varchar2(30)	N
STATE_PROVINCE	varchar2(25)	N
POSTAL_CODE	varchar2(12)	N

- SAT Table Structure:
  - SAT PK:
    - PARENT HUB / LINK PK: calculated field, HASH (SHA-265, MD5) of the Business Key Columns of the Parent HUB or LINK
    - LOAD DATETIME: keeps track of historical changes
  - HASH_DIFF: calculated field, HASH (SHA-265, MD5) of the all the descriptive columns
  - RECORD SOURCE SYSTEM: informs about the source system the link came from
  - DESCRITPIVE COLUMNS: all the required descriptive attributes of a Business Context defined by a BK

- SATs are represented in the color YELLOW

SATS

HASH_DIFF




```
CAST(SHA2(
          COALESCE(STREET_ADDRESS,'')
    || '|' || COALESCE(CITY,'')
    || '|' || COALESCE(STATE_PROVINCE,'')
    || '|' || COALESCE(POSTAL_CODE,'')
    ) AS CHAR(64)) AS ROW_HASH
```

What's so special about HASH_DIFF?

It is the hash of all the descriptive attributes (some also include the BK)

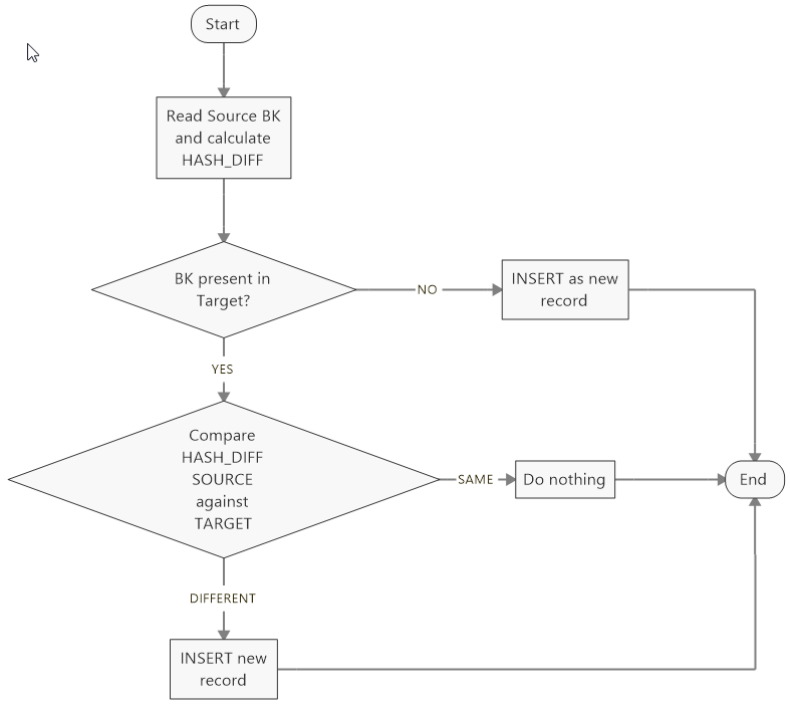
It is used to improve performance when loading new satellite rows (of which one or more attributes have changed).

The more columns the satellite has, the higher the performance gain will be.

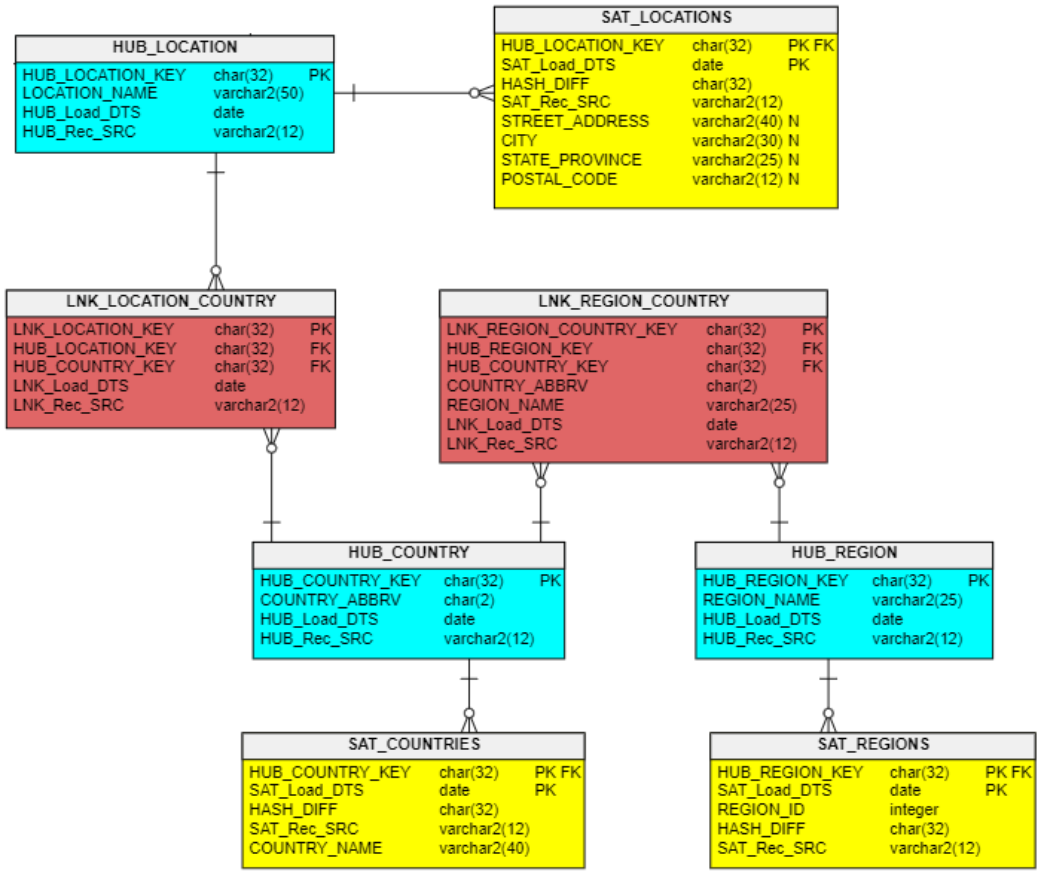
Synonyms, that describe it better:

- RowHash, RecordHash, DataHash or AttributeValuesHash

SATS LOADING



- Use of a Hash-based column for determining if a record in the source has changed from what was previously loaded into the Data Vault
- Calculate a Hash on the combination of all the descriptive columns in the Sat
- On every load, you must calculate a Hash on the inbound values (using the same formula) and then compare that to the HASH_DIFF column in the Sat of the most recent row for the same HUB (or LINK) Business Key.
- If the Hash is different, then you create a new row. If not, you do nothing (which is way faster).
- The difference with SCD2 is that instead of comparing every single column in the source feed to every single column in the SAT , you only must compare one column – the HASH_DIFF



SUMMARY

HUBs = Business Keys
HUBs make it business driven and allow for integration across different systems

LINKs = Associations / Transactions
LINKs give you the flexibility to absorb structural and business rule changes without re-engineering the existing structure and without reloading any data

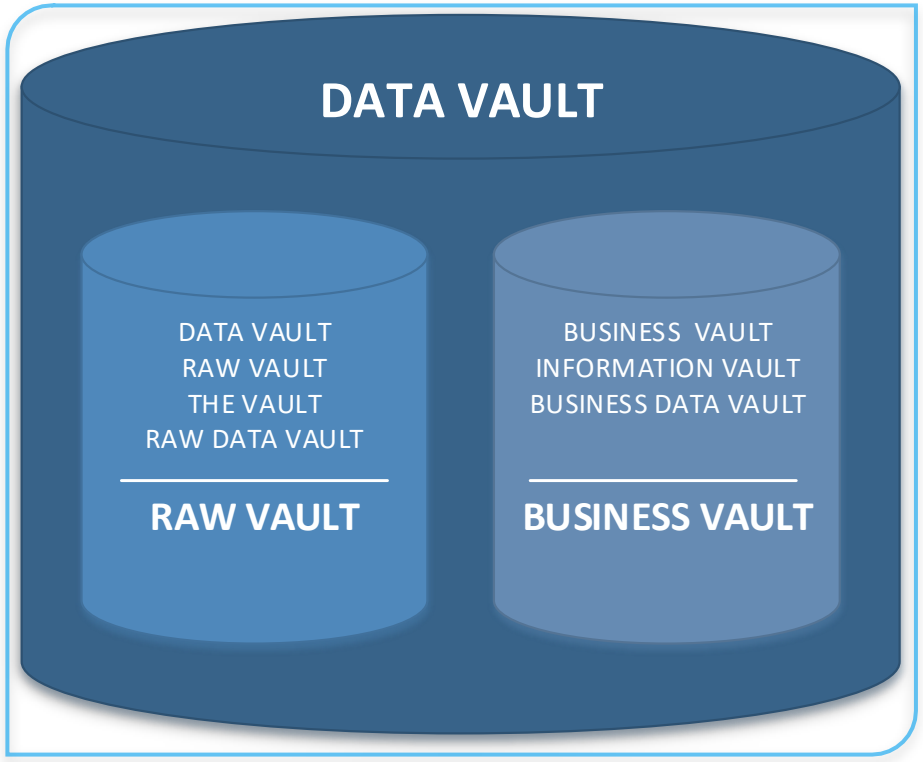
SATs = Descriptors
SATs give you the adaptability to record history , unquestionable auditability and traceability to your source systems

Agenda

Business Vault

- What
- Point-In-Time (PIT)
- Bridge Tables
- Predefined Calculations

INTRO BUSINESS VAULT



Business Vault (BV): It is an **extension** of a Raw Vault that applies **business rules, denormalizations, calculations**, and other query assistance functions in order to **facilitate user access and reporting**.

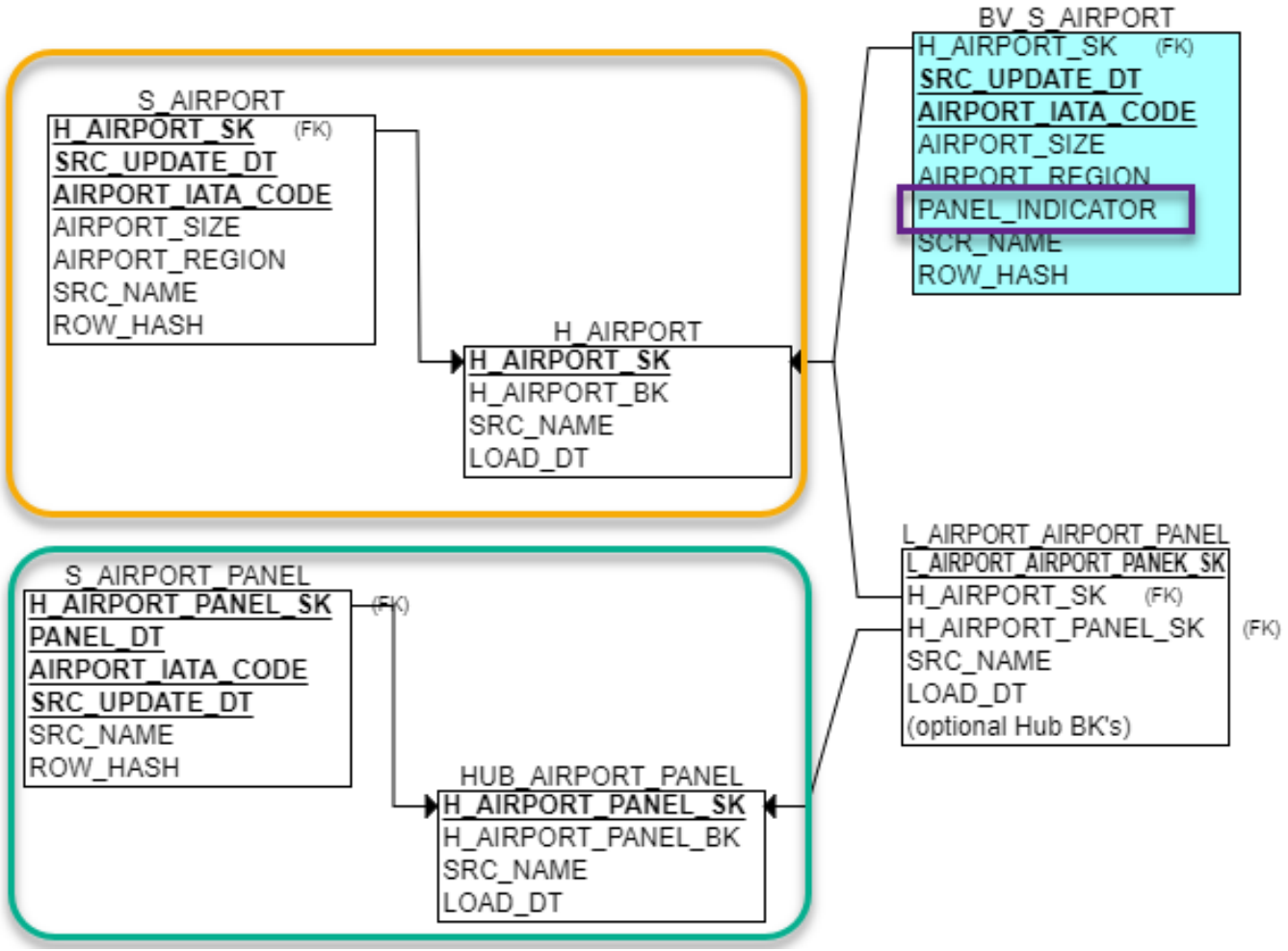
Business Vault tables should be refreshed once their dependent Raw Vault tables are refreshed

Modeled in DV Style: HUBs, LINKs, SATs

WHAT to put in A BUSINESS VAULT

Use and transform the raw data from the RAW VAULT:

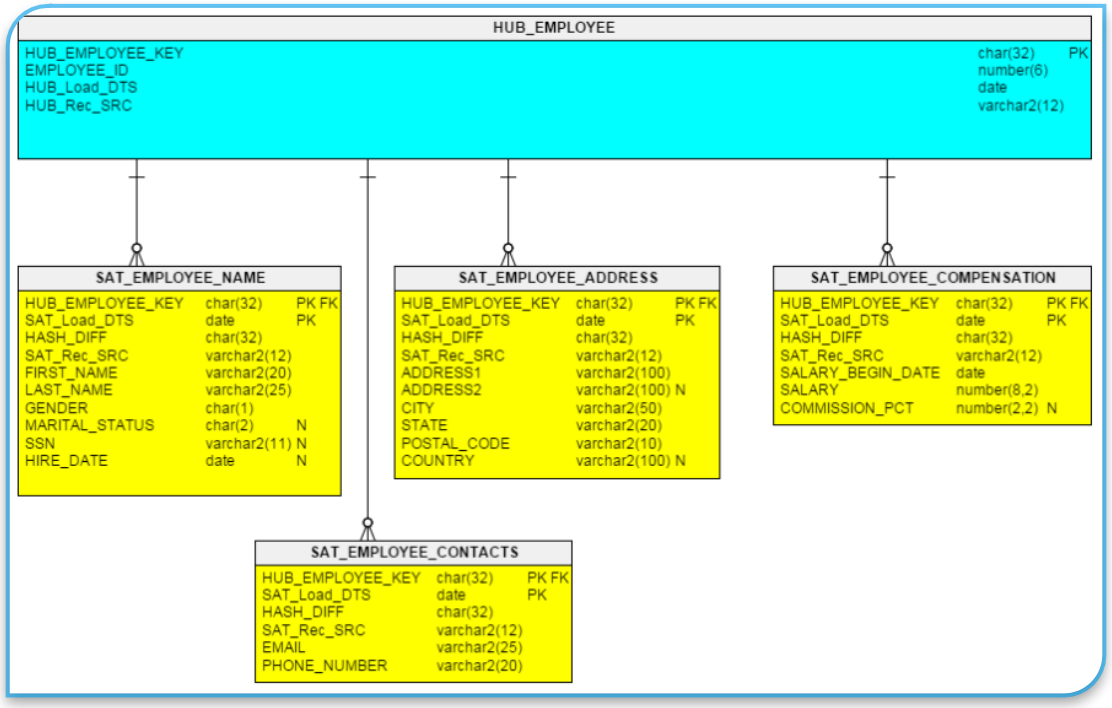
- apply rules or functions to the raw data
- Useful for most of the business users
- Guarantee the same calculation for everyone, as opposed to doing these repeatedly into different multiple marts
- Examples:
  - data cleansing
  - data quality
  - accounting rules
  - well-defined repeatable calculations (VAT, net profit, total compensation, VAT, etc.)



Example

AIRPORT context
PANEL context : list of airports we want to compare ourselves against
Create a new SAT for AIRPORT, including a PANEL_INDICATOR

Point-in-time table (PIT)



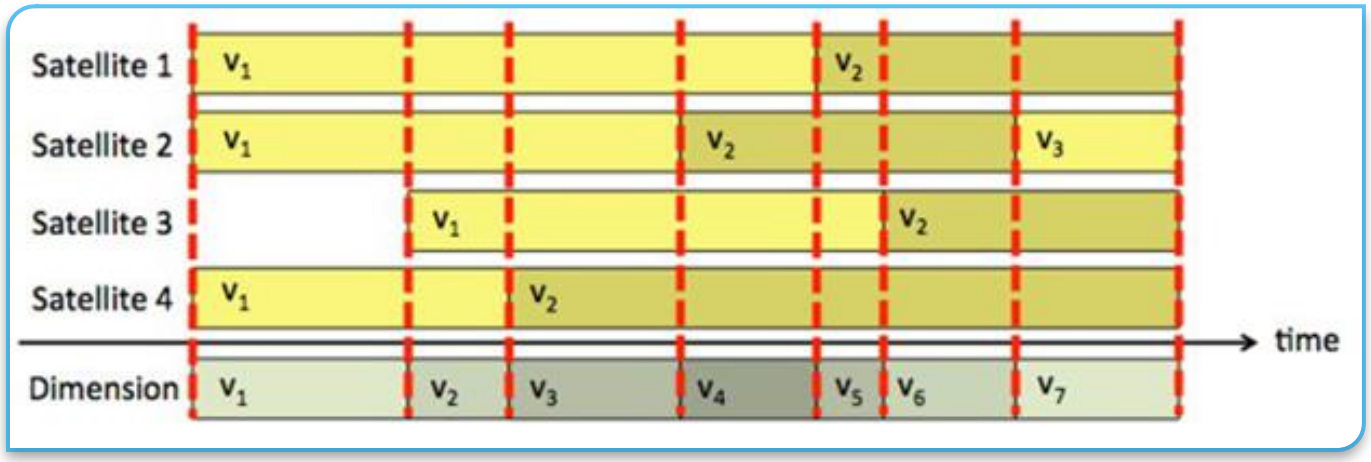
A Point-in-Time (PIT) table is a SAT that helps when we need to query data from a HUB that has multiple SATs

REMEMBER
A HUB can have multiple SATs for several reasons:

- Multiple Source Systems with each their own context and attributes
- Split Data into Contexts
- Different Change-rate of the attributes

QUERY WITHOUT PIT

Each SAT has its own TIMELINE of changes
The challenge is to figure out how to get the data out of the RAW VAULT and get a result that is consistent across time



Query example: get current version

One SAT

```
SELECT
        H.HUB_EMPLOYEE_KEY, H.EMPLOYEE_ID,
        S.FIRST_NAME, S.LAST_NAME,
        S.HIRE_DATE
FROM HUB_EMPLOYEE H

JOIN SAT_EMPLOYEE_NAME S

ON H. HUB_EMPLOYEE_KEY = S. HUB_EMPLOYEE_KEY

WHERE S.SAT_LOAD_DTS =
        (SELECT (MAX(S2.SAT_LOAD_DTS)
        FROM SAT_EMPLOYEE_NAME S2
        WHERE H. HUB_EMPLOYEE_KEY = S2.
HUB_EMPLOYEE_KEY)
```

Two SATs

```
SELECT

        H.HUB_EMPLOYEE_KEY, H.EMPLOYEE_ID, S.FIRST_NAME,
S.LAST_NAME,

        S.HIRE_DATE, C.SALARY

FROM HUB_EMPLOYEE H

JOIN SAT_EMPLOYEE_NAME S

ON H. HUB_EMPLOYEE_KEY = S. HUB_EMPLOYEE_KEY

WHERE S.SAT_LOAD_DTS =

        (SELECT (MAX(S2.SAT_LOAD_DTS)

        FROM SAT_EMPLOYEE_NAME S2

        WHERE H. HUB_EMPLOYEE_KEY = S2. HUB_EMPLOYEE_KEY)

JOIN SAT_EMPLOYEE_COMPENSATION C

ON H. HUB_EMPLOYEE_KEY = C. HUB_EMPLOYEE_KEY

WHERE C.SAT_LOAD_DTS =

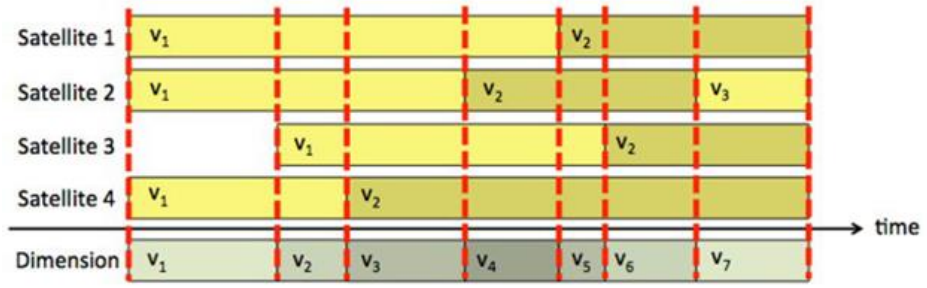
        (SELECT (MAX(S3.SAT_LOAD_DTS)

        FROM SAT_EMPLOYEE_NAME S3

        WHERE H. HUB_EMPLOYEE_KEY = S3. HUB_EMPLOYEE_KEY)
```

Query example: get historical version

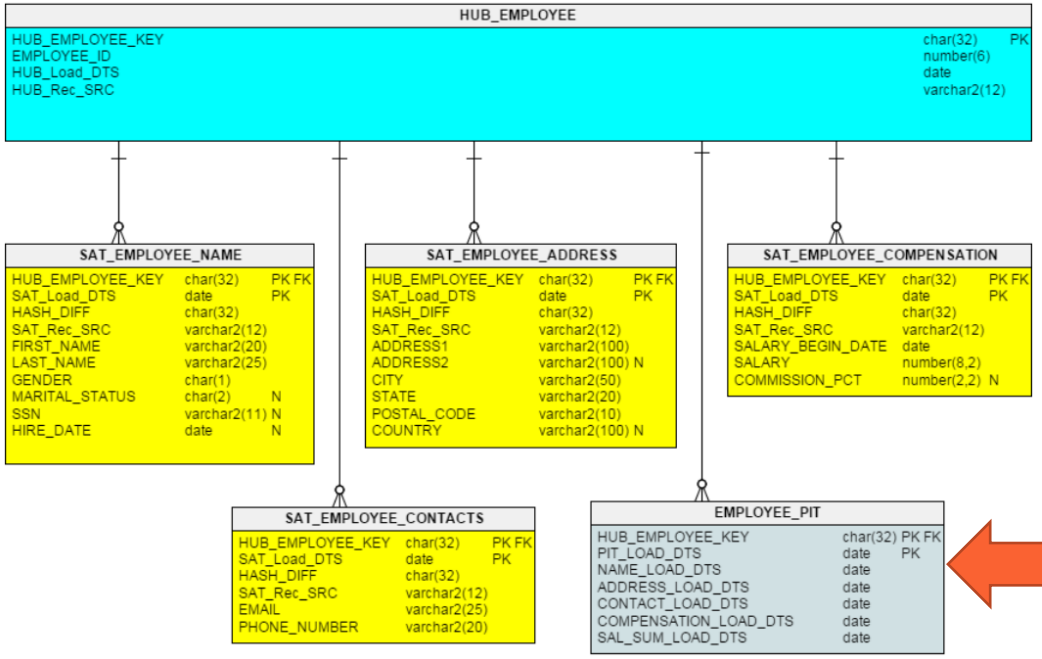
You now need to get the row of each SAT that was valid during a specific time period or on a specific date.
The SQL will become more complicated, and a source of errors:



Two SATs

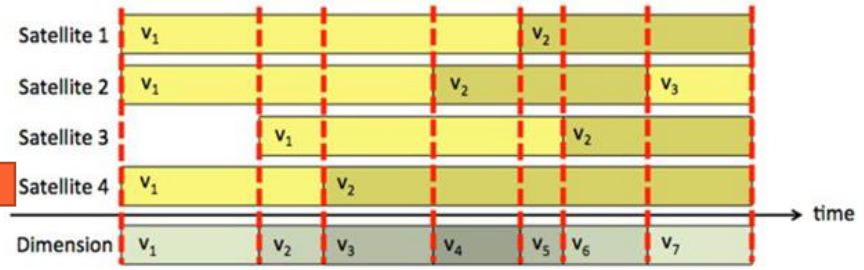
```
SELECT
        H.HUB_EMPLOYEE_KEY, H.EMPLOYEE_ID, S.FIRST_NAME, S.LAST_NAME,
        S.HIRE_DATE, C.SALARY
FROM HUB_EMPLOYEE H
JOIN SAT_EMPLOYEE_NAME S
ON H. HUB_EMPLOYEE_KEY = S. HUB_EMPLOYEE_KEY
WHERE S.SAT_LOAD_DTS =
        (SELECT (MAX(S2.SAT_LOAD_DTS)
        FROM SAT_EMPLOYEE_NAME S2
        WHERE H. HUB_EMPLOYEE_KEY = S2. HUB_EMPLOYEE_KEY
        AND S2.SAT_LOAD_DTS < '20201015')
JOIN SAT_EMPLOYEE_COMPENSATION C
ON H. HUB_EMPLOYEE_KEY = C. HUB_EMPLOYEE_KEY
WHERE C.SAT_LOAD_DTS =
        (SELECT (MAX(S3.SAT_LOAD_DTS)
        FROM SAT_EMPLOYEE_NAME S3
        WHERE H. HUB_EMPLOYEE_KEY = S3. HUB_EMPLOYEE_KEY
        AND S2.SAT_LOAD_DTS < '25-Nov-2014')
```

The solution: PIT



Load the PIT-table every day:

- For each PIT_LOAD_DT, find the current LOAD_DT of each involved SAT.



H_EMPLOYEE_SK	PIT_LOAD_DT	SAT1_DT	SAT2_DT	SAT3_DT	SAT4_DT
HASH_KEY_emp1	DT1	DT1 (V1 loaded)	DT1 (V1 loaded)	null	DT1 (V1 loaded)
HASH_KEY_emp1	DT2	DT1 (V1 loaded)	DT1 (V1 loaded)	DT2 (V1 loaded)	DT1 (V1 loaded)
HASH_KEY_emp1	DT3	DT1 (V1 loaded)	DT1 (V1 loaded)	DT2 (V1 loaded)	DT3 (V2 loaded)
HASH_KEY_emp1	DT4	DT1 (V1 loaded)	DT4 (V2 loaded)	DT2 (V1 loaded)	DT3 (V2 loaded)
HASH_KEY_emp1	DT5	DT5 (V2 loaded)	DT4 (V2 loaded)	DT2 (V1 loaded)	DT3 (V2 loaded)
HASH_KEY_emp1	DT6	DT5 (V2 loaded)	DT4 (V2 loaded)	DT6 (V2 loaded)	DT3 (V2 loaded)
HASH_KEY_emp1	DT7	DT5 (V2 loaded)	DT7 (V3 loaded)	DT6 (V2 loaded)	DT3 (V2 loaded)

TWO SCENARIOS of PIT TABLES

No GAPs in Timeline

```
SELECT
                H.HUB_EMPLOYEE_KEY, H.EMPLOYEE_ID,
                S.FIRST_NAME,S.LAST_NAME, …
FROM HUB_EMPLOYEE H
JOIN SAT_EMPLOYEE_NAME NM
ON H. HUB_EMPLOYEE_KEY = NM. HUB_EMPLOYEE_KEY
JOIN SAT_EMPLOYEE_COMPENSATION COMP
ON H. HUB_EMPLOYEE_KEY = COMP. HUB_EMPLOYEE_KEY
JOIN SAT_EMPLOYEE_ADDRESS ADDR
ON H. HUB_EMPLOYEE_KEY = ADDR. HUB_EMPLOYEE_KEY
JOIN SAT_EMPLOYEE_CONTACTS CNTC
ON H. HUB_EMPLOYEE_KEY = CNTC. HUB_EMPLOYEE_KEY
JOIN EMPLOYEE_PIT P
ON H. HUB_EMPLOYEE_KEY = P. HUB_EMPLOYEE_KEY

        AND P.PIT_LOAD_DTS = '15-FEB-2015'
        AND P.NAME_LOAD_DTS = NM.SAT_LOAD_DTS
        AND P.ADDRESS_LOAD_DTS = ADDR.SAT_LOAD_DTS
        AND P.CONTACT_LOAD_DTS = CNTC.SAT_LOAD_DTS
        AND P.COMENSATION_LOAD_DTS = COMP.SAT_LOAD_DTS)
```

GAP's in Timeline

```
…

JOIN EMPLOYEE_PIT P

ON H. HUB_EMPLOYEE_KEY = P. HUB_EMPLOYEE_KEY

AND P.PIT_LOAD_DTS =

        (SELECT MAX( P2.PIT_LOAD_DTS)

        FROM EMPLOYEE_PIT P2

        WHERE H. HUB_EMPLOYEE_KEY = P2. HUB_EMPLOYEE_KEY

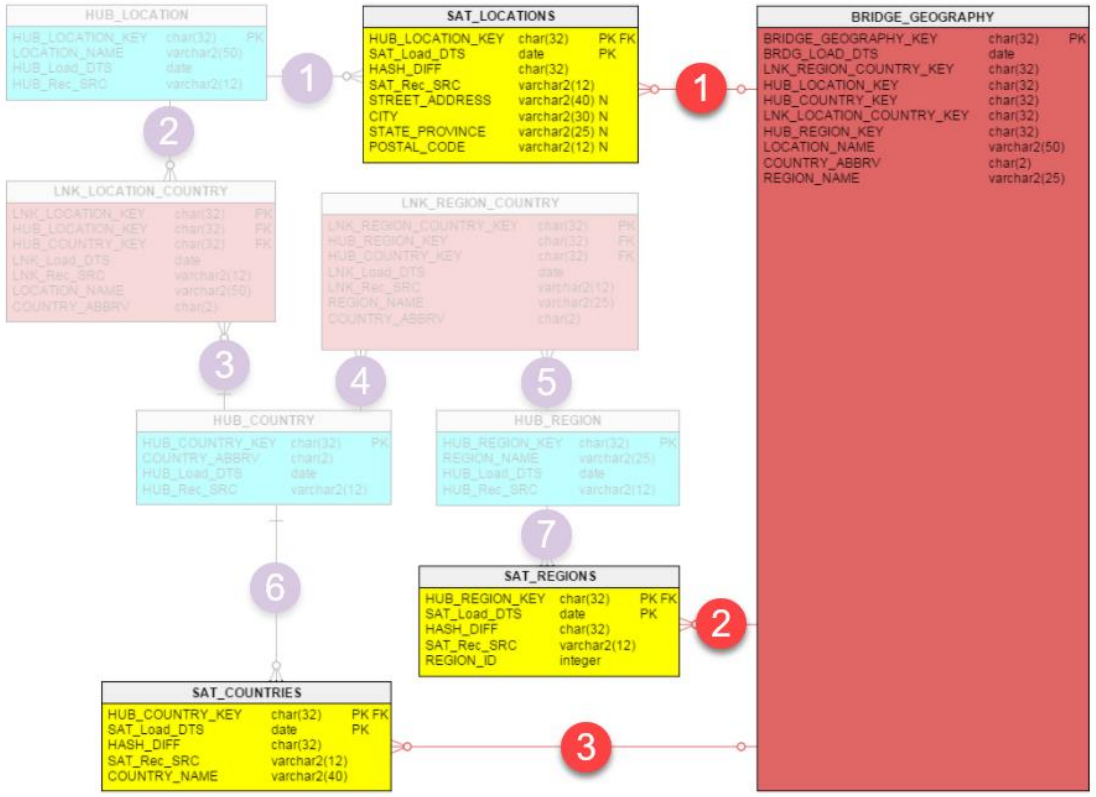
        AND P2.PIT_LOAD_DTS <= '15-FEB-2015')
AND P.NAME_LOAD_DTS = NM.SAT_LOAD_DTS

AND P.ADDRESS_LOAD_DTS = ADDR.SAT_LOAD_DTS

AND P.CONTACT_LOAD_DTS = CNTC.SAT_LOAD_DTS

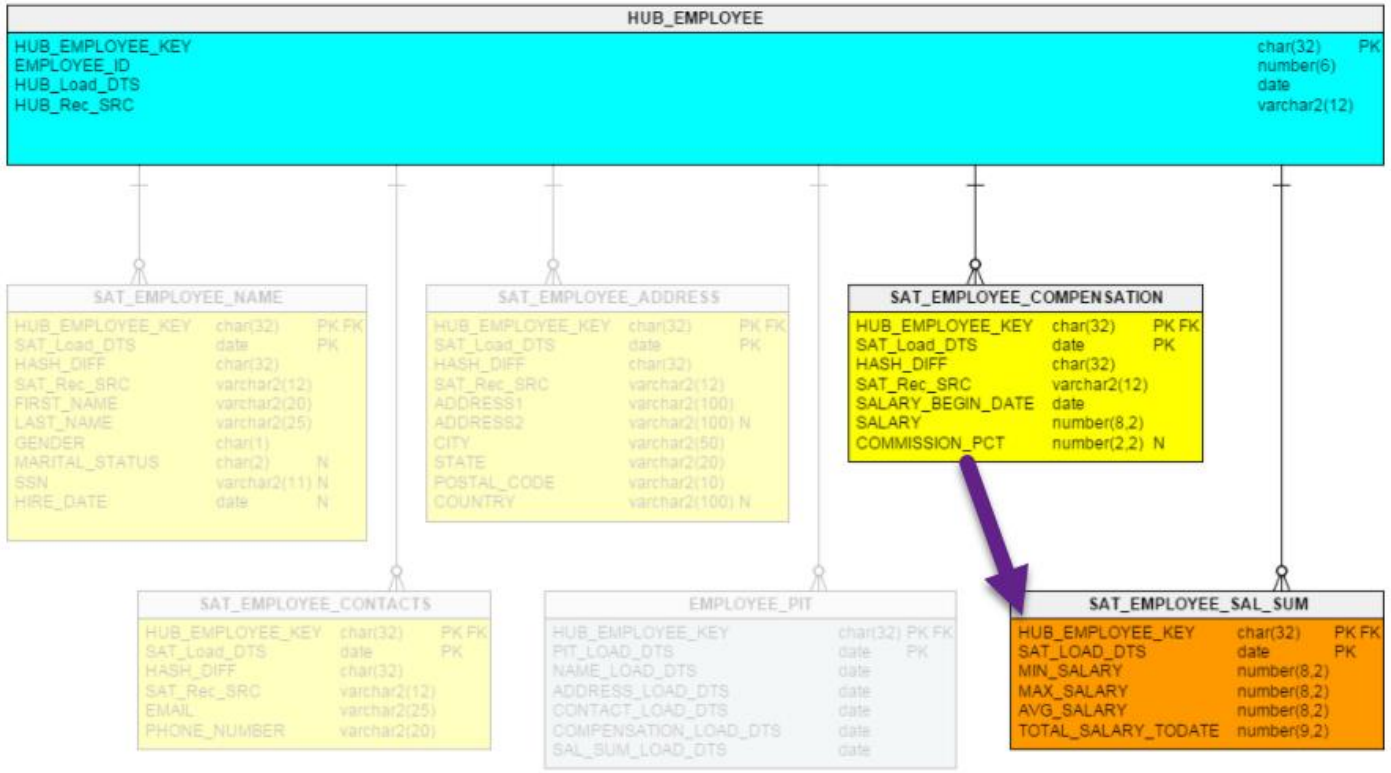
AND P.COMENSATION_LOAD_DTS = COMP.SAT_LOAD_DTS)
```

BRIDGE TABLE



- A BRIDGE TABLE is a derived LINK table, linking multiple HUBs and LINKs together
- Easier Queries
- Performance improvement of query execution
- Add Bridge tables to your Business Vault only after having encountered query performance issue

PREDEFINED CALCULATION



- LINKs or SATs attached to existing HUBs or LINKs in the Raw Vault
- Consistent pre-calculations for the Data Marts
- Driven by the business requirements

Conclusions recap

- Data Modeling technique
- Multiple and changing source systems
- Historical Data
- Normalized
- With DV, you get :
  - Agility
  - Flexibility
  - Auditability
  - Scalability
  - Speed to market
- “All the data, all of the time”
- Single version of the facts ←→ Single version of the truth
- Support parallel loading
- Very efficient load process :
  - No business rules (in BV or Data Marts)
  - No data cleansing (in BV or Data Marts)
  - Only inserts, no updates/deletes
- ETL generation
  - HUBs, LINKs and SATs can be loaded in a highly automated way
  - https://github.com/joyn/data-vault-framework

THOMAS MORE | UNIVERSITY OF APPLIED SCIENCES

Agenda

Assignment: Thomas More Airport Survey

Assignment
“Thomas More airport Survey”

Create a DV model supporting the following requirements :

Context : Airport Quality Survey at the flight departures

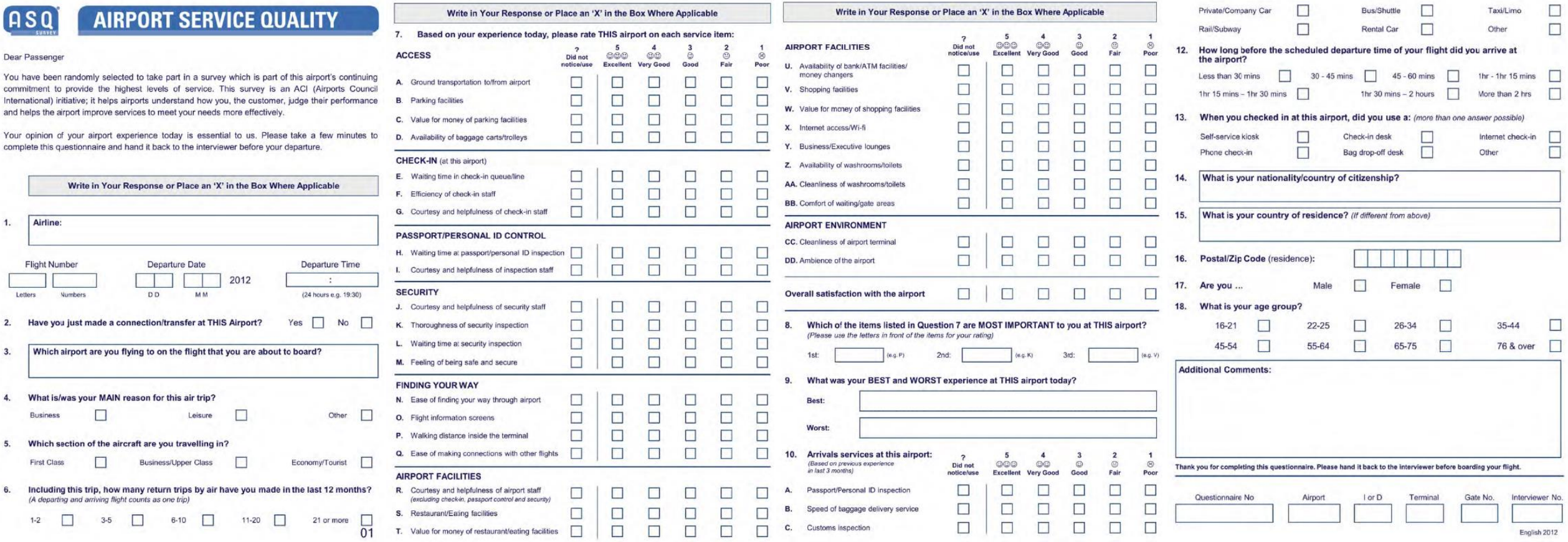
- Questionnaire:
  - Flight Info
  - Passenger Info (no personal data)
  - General Questions
  - Scored Questions

Reporting Requirements:

- Rating Evolution per Question and Category for “Thomas More Airport”
- Compare our rating against other airports (rank and position)
  - EU airports
  - Panel airports (selected list of airports in the world with comparable size)
- Alert Domain Responsible when their domain scores are bad

BONUS: provide the possibility to report the score evolution in time for Sanitary questions, by gender, age group and terminal.

ASQ Survey



ASQ SURVEY

AIRPORT SERVICE QUALITY

Dear Passenger

You have been randomly selected to take part in a survey which is part of this airport's continuing commitment to provide the highest levels of service. This survey is an ACI (Airports Council International) initiative; it helps airports understand how you, the customer, judge their performance and helps the airport improve services to meet your needs more effectively.

Your opinion of your airport experience today is essential to us. Please take a few minutes to complete this questionnaire and hand it back to the interviewer before your departure.

Write in Your Response or Place an 'X' in the Box Where Applicable

1. **Airline:**

Flight Number (Letters / Numbers) — Departure Date (D D / M M / 2012) — Departure Time (24 hours e.g. 19:30)

2. **Have you just made a connection/transfer at THIS Airport?** Yes ☐ No ☐

3. **Which airport are you flying to on the flight that you are about to board?**

4. **What is/was your MAIN reason for this air trip?**
Business ☐ Leisure ☐ Other ☐

5. **Which section of the aircraft are you travelling in?**
First Class ☐ Business/Upper Class ☐ Economy/Tourist ☐

6. **Including this trip, how many return trips by air have you made in the last 12 months?**
(A departing and arriving flight counts as one trip)
1-2 ☐ 3-5 ☐ 6-10 ☐ 11-20 ☐ 21 or more ☐

01

Write in Your Response or Place an 'X' in the Box Where Applicable

7. **Based on your experience today, please rate THIS airport on each service item:**

	? Did not notice/use	5 ☺☺☺ Excellent	4 ☺☺ Very Good	3 ☺ Good	2 😐 Fair	1 ☹ Poor
ACCESS						
A. Ground transportation to/from airport	☐	☐	☐	☐	☐	☐
B. Parking facilities	☐	☐	☐	☐	☐	☐
C. Value for money of parking facilities	☐	☐	☐	☐	☐	☐
D. Availability of baggage carts/trolleys	☐	☐	☐	☐	☐	☐
CHECK-IN (at this airport)						
E. Waiting time in check-in queue/line	☐	☐	☐	☐	☐	☐
F. Efficiency of check-in staff	☐	☐	☐	☐	☐	☐
G. Courtesy and helpfulness of check-in staff	☐	☐	☐	☐	☐	☐
PASSPORT/PERSONAL ID CONTROL						
H. Waiting time at passport/personal ID inspection	☐	☐	☐	☐	☐	☐
I. Courtesy and helpfulness of inspection staff	☐	☐	☐	☐	☐	☐
SECURITY						
J. Courtesy and helpfulness of security staff	☐	☐	☐	☐	☐	☐
K. Thoroughness of security inspection	☐	☐	☐	☐	☐	☐
L. Waiting time at security inspection	☐	☐	☐	☐	☐	☐
M. Feeling of being safe and secure	☐	☐	☐	☐	☐	☐
FINDING YOUR WAY						
N. Ease of finding your way through airport	☐	☐	☐	☐	☐	☐
O. Flight information screens	☐	☐	☐	☐	☐	☐
P. Walking distance inside the terminal	☐	☐	☐	☐	☐	☐
Q. Ease of making connections with other flights	☐	☐	☐	☐	☐	☐
AIRPORT FACILITIES						
R. Courtesy and helpfulness of airport staff *(excluding check-in, passport control and security)*	☐	☐	☐	☐	☐	☐
S. Restaurant/Eating facilities	☐	☐	☐	☐	☐	☐
T. Value for money of restaurant/eating facilities	☐	☐	☐	☐	☐	☐

Write in Your Response or Place an 'X' in the Box Where Applicable

	? Did not notice/use	5 ☺☺☺ Excellent	4 ☺☺ Very Good	3 ☺ Good	2 😐 Fair	1 ☹ Poor
AIRPORT FACILITIES						
U. Availability of bank/ATM facilities/ money changers	☐	☐	☐	☐	☐	☐
V. Shopping facilities	☐	☐	☐	☐	☐	☐
W. Value for money of shopping facilities	☐	☐	☐	☐	☐	☐
X. Internet access/Wi-fi	☐	☐	☐	☐	☐	☐
Y. Business/Executive lounges	☐	☐	☐	☐	☐	☐
Z. Availability of washrooms/toilets	☐	☐	☐	☐	☐	☐
AA. Cleanliness of washrooms/toilets	☐	☐	☐	☐	☐	☐
BB. Comfort of waiting/gate areas	☐	☐	☐	☐	☐	☐
AIRPORT ENVIRONMENT						
CC. Cleanliness of airport terminal	☐	☐	☐	☐	☐	☐
DD. Ambience of the airport	☐	☐	☐	☐	☐	☐
Overall satisfaction with the airport	☐	☐	☐	☐	☐	☐

8. **Which of the items listed in Question 7 are MOST IMPORTANT to you at THIS airport?**
(Please use the letters in front of the items for your rating)
1st: ☐ (e.g. P) 2nd: ☐ (e.g. K) 3rd: ☐ (e.g. V)

9. **What was your BEST and WORST experience at THIS airport today?**
Best:
Worst:

10. **Arrivals services at this airport:** *(Based on previous experience in last 3 months)*

	? Did not notice/use	5 ☺☺☺ Excellent	4 ☺☺ Very Good	3 ☺ Good	2 😐 Fair	1 ☹ Poor
A. Passport/Personal ID inspection	☐	☐	☐	☐	☐	☐
B. Speed of baggage delivery service	☐	☐	☐	☐	☐	☐
C. Customs inspection	☐	☐	☐	☐	☐	☐

Private/Company Car ☐ Bus/Shuttle ☐ Taxi/Limo ☐
Rail/Subway ☐ Rental Car ☐ Other ☐

12. **How long before the scheduled departure time of your flight did you arrive at the airport?**
Less than 30 mins ☐ 30 - 45 mins ☐ 45 - 60 mins ☐ 1hr - 1hr 15 mins ☐
1hr 15 mins – 1hr 30 mins ☐ 1hr 30 mins – 2 hours ☐ More than 2 hrs ☐

13. **When you checked in at this airport, did you use a:** *(more than one answer possible)*
Self-service kiosk ☐ Check-in desk ☐ Internet check-in ☐
Phone check-in ☐ Bag drop-off desk ☐ Other ☐

14. **What is your nationality/country of citizenship?**

15. **What is your country of residence?** *(If different from above)*

16. **Postal/Zip Code** (residence): ☐☐☐☐☐☐☐☐

17. **Are you ...** Male ☐ Female ☐

18. **What is your age group?**
16-21 ☐ 22-25 ☐ 26-34 ☐ 35-44 ☐
45-54 ☐ 55-64 ☐ 65-75 ☐ 76 & over ☐

Additional Comments:

Thank you for completing this questionnaire. Please hand it back to the interviewer before boarding your flight.

Questionnaire No | Airport | I or D | Terminal | Gate No. | Interviewer No.

English 2012

Source Files

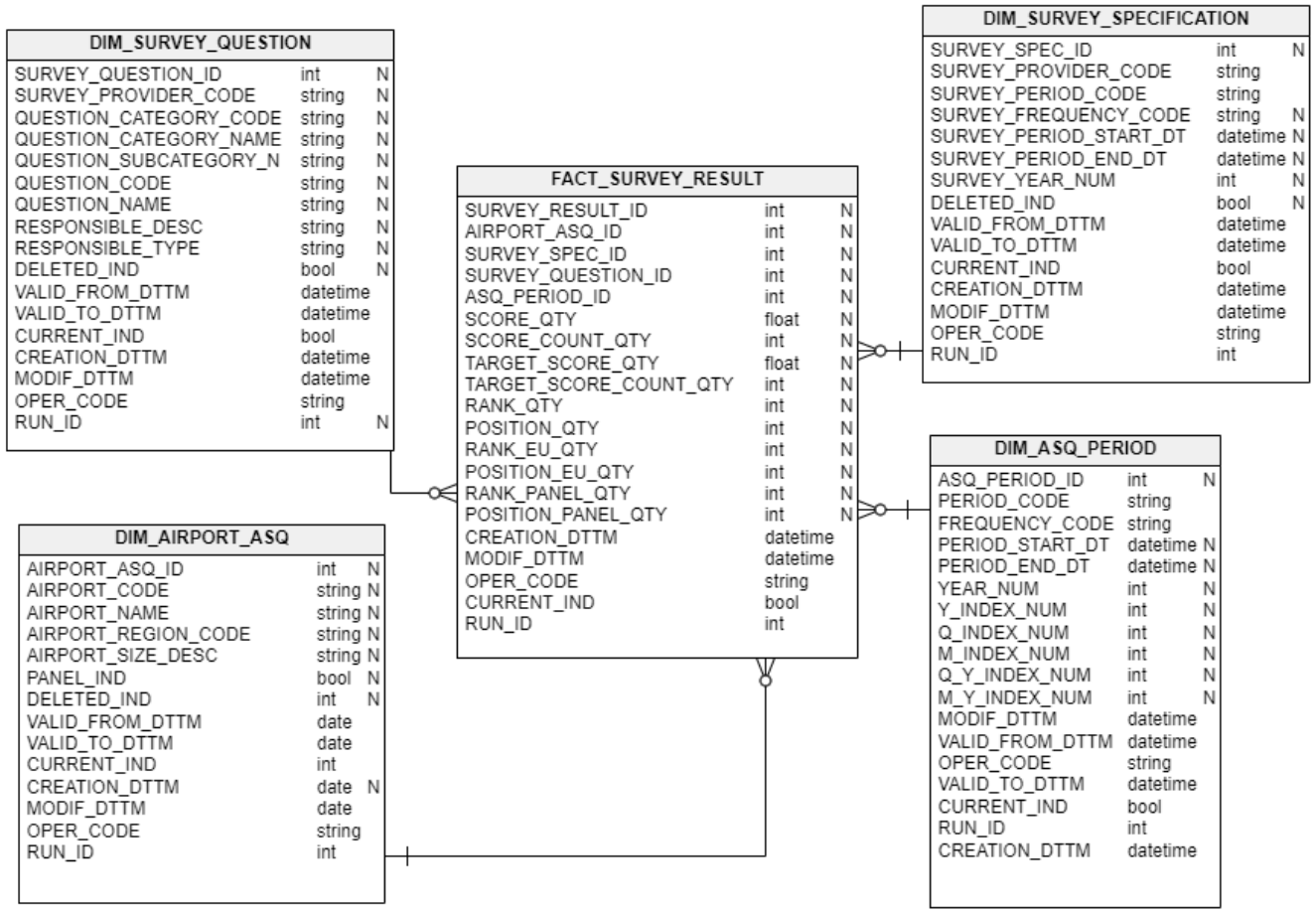
AIRPORT_SIZE	
AIRPORT_SIEZ_CODE	string
AIRPORT_SIZE_DESC	string
VALID_FROM_DT	date

AIRPORT_PANEL		
AIRPORT_CODE	varchar(50)	N

QUESTION_RESPONSIBLE		
SURVEY_PROVIDER	varchar(50)	N
VALID_FROM_DT	varchar(50)	N
CATEGORY_CODE	varchar(10)	N
CATEGORY_NAME	varchar(150)	N
SUB_CATEGORY_NAME	varchar(50)	N
QUESTION_CODE	varchar(10)	N
QUESTION_NAME	varchar(150)	N
XLS_COLUMN_NAME	varchar(100)	N
RESPONSIBLE_NAME	varchar(50)	N
RESPONSIBLE_TYPE	varchar(10)	N
DB_COLUMN_NAME	varchar(100)	N

ASQ_SURVEY		
AIRPORT	varchar(25)	N
AIRLINE	varchar(25)	N
AIRLINE_LETTER	varchar(25)	N
AIRLINE_NUMBER	varchar(25)	N
FLIGHT_DESTINATION	varchar(25)	N
DEPARTURE_TIME_HHMM	varchar(25)	N
DEPARTURE_TIME	varchar(25)	N
DEPARTURE_DAY	varchar(2)	N
DEPARTURE_MONTH	varchar(2)	N
DEPARTURE_YEAR	varchar(4)	N
DEPARTURE_DATE	varchar(25)	N
Q_CONNECT	varchar(1)	N
Q_REASON	varchar(1)	N
Q_CLASS	varchar(1)	N
Q_NBR_TRIPS	varchar(1)	N
SQ_TRANSPORT	varchar(1)	N
SQ_PARKING	varchar(1)	N
SQ_VFM_PARKING	varchar(1)	N
SQ_BAGGAGE_CARTS	varchar(1)	N
SQ_AIRLINE_WAIT	varchar(1)	N
SQ_AIRLINE_EFFIC	varchar(1)	N
SQ_AIRLINE_COURT	varchar(1)	N
SQ_WAITING_PASSPORT	varchar(1)	N
SQ_INSPECTION_STAFF	varchar(1)	N
SQ_SECURITY_STAFF	varchar(1)	N
SQ_THROUGH_INSPECT	varchar(1)	N
SQ_WAITING_INSPECT	varchar(1)	N
SQ_FEEL_SAFE	varchar(1)	N
SQ_FINDING_WAY	varchar(1)	N
SQ_SCREENS	varchar(1)	N
SQ_WALKING	varchar(1)	N
SQ_CONNECTIONS	varchar(1)	N
SQ_AIRPORT_STAFF	varchar(1)	N
SQ_RESTAURANTS	varchar(1)	N
SQ_VFM_RESTAURANT	varchar(1)	N
SQ_BANK	varchar(1)	N
SQ_SHOPPING	varchar(1)	N
SQ_VFM_SHOPPING	varchar(1)	N
SQ_COMPUTER	varchar(1)	N
SQ_AIRLINE_LOUNGE	varchar(1)	N
SQ_WASHROOMS_AV	varchar(1)	N
SQ_WASHROOMS_CLEAN	varchar(1)	N
SQ_WAITING_AREAS	varchar(1)	N
SQ_CLEANLINESS	varchar(1)	N
SQ_AMBIENCE	varchar(1)	N
SQ_OVERALL_SAT	varchar(1)	N
IMPORTANCE_1ST	varchar(10)	N
IMPORTANCE_2ND	varchar(10)	N
IMPORTANCE_3RD	varchar(10)	N
SQ_PASSPORT_ARRIVAL	varchar(1)	N
SQ_SPEED_BAG	varchar(1)	N
SQ_CUSTOMS	varchar(1)	N
Q_TRANSPORT_MODE	varchar(1)	N
Q_TIME_BEFORE_DEPRTR	varchar(1)	N
Q_CHECKIN_SELFSERVICE	varchar(1)	N
Q_CHECKIN_DESK	varchar(1)	N
Q_CHECKIN_INTERNET	varchar(1)	N
Q_CHECKIN_PHONE	varchar(1)	N
Q_CHECKIN_BAG	varchar(1)	N
Q_CHECKIN_OTHERS	varchar(1)	N
NATIONALITY	varchar(25)	N
RESIDENCY	varchar(25)	N
ZIP_CODE	varchar(25)	N
GENDER	varchar(1)	N
AGE	varchar(1)	N
TERMINAL	varchar(25)	N
GATE_NO	varchar(25)	N
SURVEY_NO_SOURCE	varchar(25)	N
SURVEY_PERIOD	varchar(4)	N
AIRPORT_SIZE	varchar(1)	N
AIRPORT_REGION	varchar(2)	N
TERMINAL_L	varchar(25)	N
FRA_T	varchar(25)	N
FRA_T1	varchar(25)	N
TRAFFIC	varchar(25)	N
WEIGHT_FACTOR	varchar(25)	N
INTERVIEWER_NUMBER	varchar(25)	N
TERMINAL_C	varchar(25)	N
ADDITIONAL_COMMENTS	varchar(4000)	N
BEST	varchar(4000)	N
WORST	varchar(4000)	N
FILE_NAME	varchar(100)	N

Datamart



FACT_SURVEY
DIM_SURVEY_SPECIFICATION
DIM_AIRPORT
DIM_SURVEY_QUESTION
DIM_PERIOD

DIM_AIRPORT

DIM_AIRPORT_ASQ		
AIRPORT_ASQ_ID	int	
AIRPORT_CODE	string	
AIRPORT_NAME	string	N
AIRPORT_REGION_CODE	string	N
AIRPORT_SIZE_DESC	string	N
PANEL_IND	bool	N
DELETED_IND	int	N
VALID_FROM_DTTM	date	
VALID_TO_DTTM	date	
CURRENT_IND	int	
CREATION_DTTM	date	N
MODIF_DTTM	date	
OPER_CODE	string	
RUN_ID	int	

- AIRPORT_SIZE_DESC: can change over time, defined in the source flat file
- PANEL_IND: flag indicating whether an airport is part of the panel for a specific period

DIM_SURVEY_QUESTION

DIM_SURVEY_QUESTION		
SURVEY_QUESTION_ID	int	
SURVEY_PROVIDER_CODE	string	N
QUESTION_CATEGORY_CODE	string	N
QUESTION_CATEGORY_NAME	string	N
QUESTION_SUBCATEGORY_N	string	N
QUESTION_CODE	string	N
QUESTION_NAME	string	N
RESPONSIBLE_DESC	string	N
RESPONSIBLE_TYPE	string	N
DELETED_IND	bool	N
VALID_FROM_DTTM	datetime	
VALID_TO_DTTM	datetime	
CURRENT_IND	bool	
CREATION_DTTM	datetime	
MODIF_DTTM	datetime	
OPER_CODE	string	
RUN_ID	int	N

- SURVEY_PROVIDER: here it's always ACI
- Question category, subcategory and code refer to the structure in the survey:

Cat	Sub Cat	Qst Code	Qst Desc
Access	7	A	ground transport from/to airport
Access	7	B	parking
Access	7	C	value for money parking
Access	7	D	baggage trolley availability

- Responsible: who, or what function in the organization (GDPR) is responsible for a question. Type can be F(unction) or D(epartment), no names.

DIM_SURVEY_SPECIFICATION

DIM_SURVEY_SPECIFICATION		
SURVEY_SPEC_ID	int	
SURVEY_PROVIDER_CODE	string	
SURVEY_PERIOD_CODE	string	
SURVEY_FREQUENCY_CODE	string	N
SURVEY_PERIOD_START_DT	datetime	N
SURVEY_PERIOD_END_DT	datetime	N
SURVEY_YEAR_NUM	int	N
DELETED_IND	bool	N
VALID_FROM_DTTM	datetime	
VALID_TO_DTTM	datetime	
CURRENT_IND	bool	
CREATION_DTTM	datetime	
MODIF_DTTM	datetime	
OPER_CODE	string	
RUN_ID	int	

- SURVEY_PROVIDER: here it's always ASQ
- SURVEY_PERIOD_CODE: YYYYQQ or YYYYMM, depending whether it's a monthly or quarterly survey
- SURVEY_FREQ_CODE: Q(uarterly) or M(onthly)

DIM_ASQ_PERIOD

DIM_ASQ_PERIOD		
ASQ_PERIOD_ID	int	
PERIOD_CODE	string	
FREQUENCY_CODE	string	
PERIOD_START_DT	datetime	N
PERIOD_END_DT	datetime	N
YEAR_NUM	int	N
Y_INDEX_NUM	int	N
Q_INDEX_NUM	int	N
M_INDEX_NUM	int	N
Q_Y_INDEX_NUM	int	N
M_Y_INDEX_NUM	int	N
MODIF_DTTM	datetime	
VALID_FROM_DTTM	datetime	
OPER_CODE	string	
VALID_TO_DTTM	datetime	
CURRENT_IND	bool	
RUN_ID	int	
CREATION_DTTM	datetime	

- Not a simple calendar, because Surveys are delivered on a monthly and quarterly basis
- PERIOD_CODE: can be YYYYMM (202010) as wel as YYYY”Q”Q (2020Q3)
- Frequency Code: M or Q
- Period start & end date: begin and end of a monthly or quarterly period

FACT_SURVEY_RESULT

FACT_SURVEY_RESULT		
SURVEY_RESULT_ID	int	
AIRPORT_ASQ_ID	int	
SURVEY_SPEC_ID	int	
SURVEY_QUESTION_ID	int	
ASQ_PERIOD_ID	int	
AVERAGE_WEIGHTED_SCORE_	float	N
SCORE_COUNT_QTY	int	N
STDEV_QTY	float	N
TARGET_SCORE_QTY	float	N
TARGET_SCORE_COUNT_QTY	int	N
TARGET_STDEV_QTY	float	N
SIGNIFICANCY_QTY	float	N
RANK_QTY	int	N
POSITION_QTY	int	N
RANK_EU_QTY	int	N
POSITION_EU_QTY	int	N
RANK_PANEL_QTY	int	N
POSITION_PANEL_QTY	int	N
CREATION_DTTM	datetime	
MODIF_DTTM	datetime	
OPER_CODE	string	
CURRENT_IND	bool	
RUN_ID	int	

- AVERAGE_WEIGHTED_SCORE and SCORE_COUNT: the “average_weighted_score” is the sum of the product of the score and the weight factor while “Score_count” is the count of the surveys, both per question, survey specification (all surveys of a period are aggregated at survey specification level), airport and date, to allow calculation averages in the reports. No AVG calculation in the fact, to avoid averages of averages in report aggregations.
- Rank and Position: two airports having the same score will have the same rank, but the position is a continuous sequence
- Target Score & Count: is the score & count of the EU Airport with rank 25